



2018. Fall.

1. Let $A \in M_{n \times n}(\mathbb{R})$ such that $A^2 = 0$. Prove

$$2 \cdot \text{rank } A \leq n.$$

Proof. Given $x \in \mathbb{R}^n$, $A(Ax) = A^2x = 0$, so $\text{Im } A \subseteq \ker A$. So, $\dim \text{Im } A \leq \dim \ker A$.
By the dimension theorem,

$$n = \text{nullity } A + \text{rank } A \geq 2 \text{rank } A \quad \square$$

2. Determine over $\mathbb{R}$,

$$A = \begin{pmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix}$$

is diagonalizable.

Proof. The characteristic polynomial of A is

$$\det \begin{pmatrix} t & 0 & 2 \\ -1 & t-2 & -1 \\ -1 & 0 & t-3 \end{pmatrix} = t(t-2)(t-3) + 2 \cdot (-1)(t-2) = (t-2)(t^2 - 3t + 2) = (t-1)(t-2)(t-2).$$

By the Cayley-Hamilton theorem, the minimal polynomial has same roots to characteristic polynomial, so we need to check that $(A-I)(A-2I) = 0$.

$$\begin{pmatrix} -1 & 0 & -2 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} -2 & 0 & -2 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = 0$$

So, the minimal polynomial of A is $(t-1)(t-2)$, it implies A is diagonalizable, because the minimal polynomial splits over $\mathbb{R}$ and separable. Explicitly,

$$\begin{pmatrix} -1 & 0 & -2 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \Leftrightarrow \begin{cases} x = -2z \\ x+y+z=0 \end{cases} \Rightarrow \ker(A-I) = \text{span} \left\{ \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} \right\}$$
$$\begin{pmatrix} -2 & 0 & -2 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \Leftrightarrow \begin{cases} x+z=0 \end{cases} \Rightarrow \ker(A-2I) = \text{span} \left\{ \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \right\}$$

Therefore, $A = UDU^{-1}$ where $U = \begin{pmatrix} -2 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & -1 \end{pmatrix}$ and $D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$.

2018, Fall,

3. Let R be a commutative ring and let $x \in R$ such that $x^{2018} = 0$.
prove that $1+x$ is unit.

Proof. observe that

$$\begin{aligned} & (1+x) \cdot (-x^{2017} + x^{2016} - x^{2015} + \dots - x + 1) \\ &= 1 - x + x^2 - \dots + x^{2016} - x^{2017} \\ & \quad + x - x^2 + \dots - x^{2016} + x^{2017} - x^{2018} \\ &= 1, \quad \text{since } x^{2018} = 0. \quad \text{So, } 1-x+\dots+x^{2016}-x^{2017} \text{ is inverse of } 1+x. \end{aligned}$$

4. prove that A_4 does not have a subgroup of order 6.

Proof. Since $|S_4:A_4| = 2$, $|A_4| = 4 \cdot 3 = 12$.

Suppose that A_4 has a subgroup H of order 6.

Then, $|A_4:H| = 12/6 = 2$, so $gH = Hg$ for any $g \in A_4$, it means $H \trianglelefteq A_4$ is normal.
Consider the afforded Homomorphism of A_4 acts A_4/H as left multiplication:

$$\mathcal{Q}: A_4 \rightarrow S_{A_4/H}: g \mapsto (x) \mapsto gxH$$

And, since every 3-cycle $\sigma \in A_4$ satisfies

$$(\mathcal{Q}(\sigma))^3 = \mathcal{Q}(\sigma^3) = \mathcal{Q}(\text{id}) = \text{id}$$

and $S_{A_4/H} \cong \mathbb{Z}/2\mathbb{Z}$ so $\mathcal{Q}(\sigma) = \text{id}$. It implies all 3-cycles are contained in $\ker \mathcal{Q}$. Meanwhile, observe that $(1\ 2\ 3), (1\ 2\ 4) \in A_4$ satisfies

$$(1\ 2\ 3)^{-1}(1\ 2\ 4)(1\ 2\ 3) = (1\ 3\ 2)(1\ 2\ 4)(1\ 2\ 3) = (1\ 4\ 3)$$

$(1\ 2\ 3)(1\ 4\ 3) = (1\ 4)(2\ 3)$, ... in this way, A_4 is generated by all 3-cycles in A_4 , so $\ker \mathcal{Q} = A_4$. By the first isomorphism theorem, $A_4/\ker \mathcal{Q} \cong \text{Im } \mathcal{Q}$ is trivial, so the action is trivial.

But, the left coset action is transitive, it cannot be trivial, contradiction. $\square$

Similar Problems,

1. Show that A_5 has no subgroup of order 15.

Proof. $|A_5| = |S_5|/2 = 60 = 2^2 \cdot 3 \cdot 5$.

Suppose that there is a subgroup $H < A_5$ of order 15.

By the Sylow Theorem, $\# \text{Syl}_3(H) \in \{1\}$ and $\# \text{Syl}_5(H) \in \{1\}$, that is,

there are order 3 subgroup $P < H$ and order 5 subgroup $Q < H$,

and both are normal subgroup of H . Since $P \cong C_3$ and $Q \cong C_5$ both are

cyclic, let $x \in P$ and $y \in Q$ such that $P = \langle x \rangle$ and $Q = \langle y \rangle$.

Consider the afforded Homomorphism by H acts P by conjugation

$$\varphi: H \rightarrow \text{Sym}(P) \cong \text{Aut}(P) \quad (h \mapsto h x h^{-1})$$

Then, $\text{Ker } \varphi = \{h \in H : h x h^{-1} = x \text{ for all } x \in P\} = C_H(P)$ and

$H / \text{Ker } \varphi \cong \text{Im } \varphi \leq \text{Aut}(P)$. But, since $\text{Aut}(P) \cong \mathbb{Z}/2\mathbb{Z}$,

and $|H|$ has no factor 2, odd number, so $|\text{Ker } \varphi| = |H|$, that is,

$C_H(P) = H$. Now, x and y commute, therefore $|xy| = 15$.

But, A_5 cannot have order 15 element, because:

order of permutation in S_5 is determined by the least common multiplier of disjoint cycle decompositions, but $\text{lcm } 5 = 5$,

$\text{lcm}(4, 1) = 4$, $\text{lcm}(3, 2) = 6$, $\text{lcm}(2, 1, 1, 1) = 2$, ...

the lcm cannot be 15, so H contradicts.

[Similar problem

Let $G = S_3 \times C_2$.

1. Find all normal subgroups of G .

2. Prove that G has no normal subgroup of order 4.

Proof. Define three maps:

$$\mathcal{Q}_1: G \rightarrow S_3: (\sigma, \alpha) \mapsto \sigma$$

$$\mathcal{Q}_2: G \rightarrow C_2: (\sigma, \alpha) \mapsto \alpha$$

$$\mathcal{Q}_3: G \rightarrow \{\pm 1\}: (\sigma, \alpha) \mapsto \text{sgn } \sigma \cdot \pi(\alpha) \quad \text{where } C_2 \stackrel{\pi}{\cong} \{\pm 1\}.$$

$$\mathcal{Q}_4: G \rightarrow \{\pm 1\} \times C_2: (\sigma, \alpha) \mapsto (\text{sgn } \sigma, \alpha).$$

$$\mathcal{Q}_5: G \rightarrow \{\pm 1\}: (\sigma, \alpha) \mapsto \text{sgn } \sigma$$

Then, $\mathcal{Q}_1$ and $\mathcal{Q}_2$ are Homomorphisms, being the Coordinatively Projection, and $\mathcal{Q}_3$ is also Homomorphism, because sgn , π are Homomorphism and $\{\pm 1\}$ is abelian. Precisely,

$$\begin{aligned} \mathcal{Q}_3((\sigma_1, \alpha_1) \cdot (\sigma_2, \alpha_2)) &= \mathcal{Q}_3(\sigma_1 \sigma_2, \alpha_1 \alpha_2) = \text{sgn } \sigma_1 \text{sgn } \sigma_2 \pi(\alpha_1) \pi(\alpha_2) \\ &= \text{sgn } \sigma_1 \pi(\alpha_1) \text{sgn } \sigma_2 \pi(\alpha_2) = \mathcal{Q}_3(\sigma_1, \alpha_1) \cdot \mathcal{Q}_3(\sigma_2, \alpha_2). \end{aligned}$$

And, $\mathcal{Q}_4$ is Homomorphism, being entrywise Homomorphism, and $\mathcal{Q}_5$ is Homomorphism, being Composition of Projection and sgn .

Now, the kernel

$$\ker \mathcal{Q}_1 = \{\text{id}\} \times C_2, \quad \ker \mathcal{Q}_2 = S_3 \times \{x^0\}, \quad \ker \mathcal{Q}_3 = A_3 \times \{x^0\} \cup (S_3 \setminus A_3) \times \{x^1\}$$

$$\ker \mathcal{Q}_4 = A_3 \times \{x^0\}, \quad \ker \mathcal{Q}_5 = A_3 \times C_2.$$

are we desired.

Are these all possibilities?

Observe that: if $N \subseteq G_1 \times G_2$ is normal, then the image of projections $\pi_i: G_1 \times G_2 \rightarrow G_i: (g_1, g_2) \mapsto g_i$, $\pi_i[N]$ is also normal of $\pi_i[G_1 \times G_2] = G_i$, because image of normal subgroup is normal in the Range, and the projections are onto. But, as we know, normal subgroup of S_3 is only $\{\text{id}\}$, A_3 , S_3 and normal subgroup of $\{\pm 1\}$ is $\{1\}$ and $\{\pm 1\}$.

2018. Fall.

7. Consider the identity $\frac{e^{-x}}{1-x} = \sum_{n=0}^{\infty} C_n x^n$.

a). Determine the sequence $(C_n) \subseteq \mathbb{R}$ and the range of $x \in \mathbb{R}$ for the identity holds.

b). Show that for each $n \geq 0$,

$$\sum_{k=0}^n \frac{C_k}{(n-k)!} = 1.$$

Proof. a),

Notice that $e^{-x} = \sum_{n=0}^{\infty} \frac{(-x)^n}{n!}$ for all $x \in \mathbb{R}$ and

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \text{ for all } |x| < 1.$$

Therefore, the coefficient C_n is determined by the Cauchy product

$$C_n = \sum_{k=0}^n \frac{(-1)^{n-k}}{(n-k)!} \cdot 1^k = \sum_{k=0}^n \frac{(-1)^{n-k}}{(n-k)!}.$$

And, the identity valid in $\mathbb{R} \cap (-1, 1) = (-1, 1)$. $\perp 5m$.

b). Since

$$\frac{1}{1-x} = e^{-x} \cdot \sum_{n=0}^{\infty} C_n x^n, \text{ so we can write for } x \in (-1, 1)$$

$$\sum_{n=0}^{\infty} x^n = \sum_{n=0}^{\infty} \sum_{k=0}^n \frac{C_k}{(n-k)!} x^n.$$

And, the coefficients equals for each $n \geq 0$, because these are infinitely differentiable, so the coefficients of power series are determined uniquely by the Taylor Theorem. Thus,

$$1 = \sum_{k=0}^n \frac{C_k}{(n-k)!} \text{ for all } n \geq 0. \perp 4m$$

Tzola, Spring

6. For each $n \in \mathbb{N}$, let

$$f_n(x) = \frac{|x|^n}{1+|x|^n}, \quad x \in \mathbb{R}.$$

a). Show that $(f_n)_{n \geq 1}$ does not converge uniformly.

b) For each fixed $q > 1$, Show that (f_n) converges uniformly on $\{x \in \mathbb{R} \mid |x| \geq q\}$ and $\{x \in \mathbb{R} \mid |x| \leq \frac{1}{q}\}$.

a). Given $x \in \mathbb{R}$,

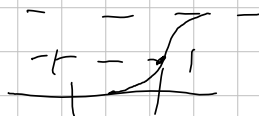
if $|x| < 1$, then $f_n(x) \rightarrow 0$ as $n \rightarrow \infty$, since $|x|^n \rightarrow 0$.

if $|x| = 1$, then $f_n(x) = \frac{1}{2}$ for all $n \in \mathbb{N}$, since $|x|^n = 1$.

if $|x| > 1$, then $f_n(x) = \frac{1}{1/|x|^n + 1} \rightarrow 1$ as $n \rightarrow \infty$, since $1/|x|^n \rightarrow 0$ as $n \rightarrow \infty$.

Therefore, $f_n \rightarrow f$ pointwise where

$$f(x) = \begin{cases} 0 & |x| < 1 \\ \frac{1}{2} & |x| = 1 \\ 1 & |x| > 1 \end{cases}$$



But, for $\epsilon = 1/4$, and for all $n \in \mathbb{N}$,

$$\sup_{0 \leq x < 1} |f_n(x) - f(x)| = \sup_{0 \leq x < 1} |f_n(x)| = \sup_{0 \leq x < 1} \frac{|x|^n}{1+|x|^n} \geq \frac{1}{4}$$

because given $n \in \mathbb{N}$, we can take $x \in [0, 1)$ such that $\frac{x^n}{1+x^n} \geq \frac{1}{4}$

Since $\lim_{x \rightarrow 1^-} \frac{|x|^n}{1+|x|^n} = \frac{1}{2}$, so, $\sup_{x \in \mathbb{R}} |f_n - f| \geq \sup_{0 \leq x < 1} |f_n - f|$ implies

$f_n \rightarrow f$ but not uniformly on $\mathbb{R}$.

b). If $|x| \geq q > 1$, then $f(x) = 1$. So, enough to show that $\sup |f_n(x) - f(x)| \rightarrow 0$.

Since $\sup_{|x| \geq q} |f_n(x) - f(x)| = \sup_{|x| \geq q} \left| \frac{1}{1+|x|^n} \right| \leq \frac{1}{1+q^n} \rightarrow 0$ as $n \rightarrow \infty$, so $f_n \rightarrow f$ uniformly on $|x| \geq q$. Similarly, for $|x| \leq \frac{1}{q}$, $f(x) = 0$ and so

$$\sup_{|x| \leq \frac{1}{q}} |f_n(x) - f(x)| = \sup_{|x| \leq \frac{1}{q}} \left| \frac{|x|^n}{1+|x|^n} \right| \leq \frac{1}{q^n + 1} \rightarrow 0 \text{ as } n \rightarrow \infty,$$

so proved all cases. $\square$

2019, Spring.

17, Let $S = \{(x, y, z) \in \mathbb{R}^3 \mid z = x \cdot f(\frac{y}{x}), x \neq 0\}$ where $f: \mathbb{R} \rightarrow \mathbb{R}$ is differentiable. Show that the tangent plane at any point of S passes through the origin. Prove by the definition of S , given $(x_0, y_0, z_0) \in S$, a map

$X: \mathcal{O} \rightarrow S: (u, v) \mapsto (u, v, u \cdot f(\frac{v}{u}))$ where $\mathcal{O}$ is an open set containing (x_0, y_0) is a parametrization of S at (x_0, y_0, z_0) . The differential of X at (x_0, y_0) is:

$$dX = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ f(\frac{v}{u}) - \frac{v}{u} \cdot f'(\frac{v}{u}) & f'(\frac{v}{u}) \end{bmatrix} \bigg|_{(u,v)=(x_0,y_0)}$$

So, the tangent plane of S at (x_0, y_0) is generated by

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ f(\frac{y_0}{x_0}) - \frac{y_0}{x_0} \cdot f'(\frac{y_0}{x_0}) \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ f'(\frac{y_0}{x_0}) \end{pmatrix} \right\}.$$

Now, for $t_1, t_2 \in \mathbb{R}$, the coordinate of a point of the tangent plane can be described by

$$(x_0, y_0, x_0 \cdot f(\frac{y_0}{x_0})) + (t_1, 0, t_1 \cdot f(\frac{y_0}{x_0}) - t_1 \cdot \frac{y_0}{x_0} f'(\frac{y_0}{x_0})) \\ + (0, t_2, t_2 \cdot f'(\frac{y_0}{x_0}))$$

is zero when $t_1 = -x_0$ and $t_2 = -y_0$, so proved $\square$

201a, Fall.

8. Let $(b_n)_{n \in \mathbb{N}}$ be a decreasing sequence of positive real numbers with $b_n \rightarrow 0$. Let $(a_n)_{n \in \mathbb{N}}$ be a sequence of complex numbers such that $(\sum_{n=1}^N a_n)_{N \in \mathbb{N}}$ is a bounded sequence.

a). Show that $\sum_{n=1}^{\infty} a_n b_n$ converges.

b). Show $\sum_{n=2}^{\infty} \frac{\sin(n\pi x)}{\log n}$ converges for all $x \in [-\pi, \pi]$.

Proof. Denote $A_n = \sum_{k=1}^n a_k$ and $A_0 = 0$. And let $M \geq 0$ be a bound for $\sum a_n$.

$$\begin{aligned} \sum_{n=1}^N a_n b_n &= \sum_{n=1}^N [A_n - A_{n+1}] \cdot b_n = \sum_{n=1}^N A_n b_n - \sum_{n=1}^N A_{n+1} b_n \left(\sum_{n=1}^N A_n b_n = \sum_{n=0}^{N-1} A_n b_{n+1} \right) \\ &= \sum_{n=1}^N A_n b_n - \left(\sum_{n=1}^N A_n b_{n+1} + A_0 b_1 - A_N b_{N+1} \right) \\ &= \sum_{n=1}^N A_n (b_n - b_{n+1}) + A_N b_{N+1}. \end{aligned}$$

Therefore, for any $m > n \geq 1$,

$$\sum_{k=n}^m a_k b_k = \sum_{k=n}^m A_k (b_k - b_{k+1}) + A_m b_{m+1} - A_n b_n$$

Now, using the hypothesis,

$$\begin{aligned} \left| \sum_{k=n}^m a_k b_k \right| &\leq \sum_{k=n}^m |A_k| \cdot |b_k - b_{k+1}| + |A_m| |b_{m+1}| + |A_n| |b_n| \\ &\leq M \cdot \left(\sum_{k=n}^m |b_k - b_{k+1}| + |b_{m+1}| + |b_n| \right) \\ &= M \cdot \left(\sum_{k=n}^m (b_k - b_{k+1}) + b_{m+1} + b_n \right) \end{aligned}$$

$$= M \cdot 2b_n \rightarrow 0 \text{ as } n \rightarrow \infty, \text{ so proved.}$$

b). Since $1/\log n$ is decreasing and positive real numbers, so by a), enough to show that $|\sum \sin(n\pi x)|$ is bounded for all $x \in [-\pi, \pi]$.

observe that: $2 \sin x \cdot \sin nx = \cos(nx - x) - \cos(nx + x)$

Therefore, $2 \sin x \cdot \sum_{n=1}^N \sin(n\pi x) = \sum_{n=1}^N [\cos((n-1)x) - \cos((n+1)x)] \leq |\cos 0| + |\cos x| + |\cos Nx| + |\cos(N+1)x| \leq 4$

so for $x \in [-\pi, \pi] \setminus \{0, \pm\pi\}$, $\left| \sum_{n=2}^N \sin n\pi x \right| \leq \frac{2}{|\sin x|}$ for any $N \in \mathbb{N}$, bounded.
for $x = 0$ or $\pm\pi$, then the sum is just zero. $\square$

2019, Fall,

1. Let $\varphi: [0, 1] \rightarrow \mathbb{R}$ be a continuous map. Evaluate

$$\lim_{\varepsilon \rightarrow 0} \int_0^1 \left[\frac{\varphi(x)}{x - i\varepsilon} - \frac{\varphi(x)}{x + i\varepsilon} \right] dx.$$

Proof. Observe that

$$\frac{1}{x - i\varepsilon} - \frac{1}{x + i\varepsilon} = \frac{2i\varepsilon}{x^2 + \varepsilon^2}$$

2022, spring.

1). Let G be a group of order 2021,

a). Show that G is not simple.

b). Show that G is cyclic.

Proof. Notice that $2021 = 43 \cdot 47$, (Using $[H, K] \leq H \cap K$ if $H, K \triangleleft G$)

2022 Spring

2. Prove or disprove that each of following is PID:

a) $\mathbb{R}[x]$

b) $\mathbb{R}[x, y]$

c) $\mathbb{Z}[\sqrt{-1}]$

d) $\mathbb{Z}[x]$

Proof.

a) $\mathbb{R}[x]$ is PID, because since $\mathbb{R}$ is a field, so $\mathbb{R}[x]$ is Euclidean domain, so for any ideal $I \subseteq \mathbb{R}[x]$, we can find a generator of I

b) $\mathbb{R}[x, y]$ is not P.I.D., because: let $(x, y) \subseteq \mathbb{R}[x, y]$.

If $(x, y) = (p(x, y))$ for some $p \in \mathbb{R}[x, y]$,

then $p(0, 0) = 0$, that is, p has zero constant term.

But, $x, y \in (x, y) = (p(x, y))$ implies $\deg_x p, \deg_y p \leq 1$,

2020. Spring.

6. Find the general solution of the differential equation

$$y'' - 3y' + 2y = \frac{1}{1+e^{-t}}.$$

Proof. Since the characteristic of $y'' - 3y' + 2y = 0$ is $r^2 - 3r + 2 = (r-1)(r-2)$, therefore the general solution of the homogeneous part is:

$$y_h(t) = C_1 e^t + C_2 e^{2t} \quad (C_1, C_2 \in \mathbb{R}).$$

And, to find a solution of the given equation, we need to find the particular solution $y_p(t)$. First, calculate the Wronskian,

$$\begin{vmatrix} e^t & e^{2t} \\ e^t & 2e^{2t} \end{vmatrix} = 2e^{3t} - e^{3t} = e^{3t}.$$

Therefore, the particular solution is

$$\begin{aligned} y_p(t) &= \int_{x_0}^x \begin{vmatrix} e^t & e^{2t} \\ e^{2t} & e^{2t} \end{vmatrix} \cdot \frac{1}{1+e^{-t}} \cdot \frac{1}{e^{3t}} dt \\ &= \int_{x_0}^x \frac{e^{t+2t} - e^{2t+t}}{e^{3t} + e^{2t}} dt \\ &= e^{2x} \int_{x_0}^x \frac{1}{e^{2t} + e^t} dt - e^x \int_{x_0}^x \frac{1}{e^t + 1} dt \end{aligned}$$

Meanwhile, since $\int \frac{1}{e^t + 1} dt = \int \frac{e^{-t}}{1 + e^{-t}} dt = -\log(1 + e^{-x})$

and $\int \frac{1}{e^{2t} + e^t} dt = \int \frac{e^{-t}}{e^t + 1} dt = \int \frac{-u}{\frac{1}{u} + 1} \cdot \frac{1}{u} du = \int \frac{-u}{1+u} du$

$$= \int -1 + \frac{1}{1+u} du = -u + \log(1+u) = -e^{-x} + \log(1+e^{-x})$$

Finally, the general solution is:

$$\begin{aligned} y_p + C_1 e_1 + C_2 e_2 &= \left[e^x + e^{2x} \log(1+e^{-x}) + e^x \cdot \log(1+e^x) \right] \\ &+ C_1 e^x + C_2 e^{2x}, \quad C_1, C_2, C_3 \in \mathbb{R}, \end{aligned}$$

12/20. Spring

15. Let $r=r(\theta)$ be a curve in $\mathbb{R}^2$ defined in polar coordinates (r, θ) with $r > 0$, $0 \leq \theta < 2\pi$.

a). Express the arc length of the curve using $r(\theta)$, $r'(\theta)$.

b). Express the curvature at $r(\theta)$ using $r(\theta)$, $r'(\theta)$, $r''(\theta)$.

Proof. Denote $x = r \cdot \cos \theta$ and $y = r \cdot \sin \theta$.

a), $\frac{dx}{d\theta} = r' \cos \theta - r \sin \theta$, $\frac{dy}{d\theta} = r' \sin \theta + r \cos \theta$.

therefore, $\int_0^{2\pi} \left| \frac{dx}{d\theta} \right| d\theta = \int_0^{2\pi} \sqrt{r'^2 + r^2} d\theta$.

b).

12/20, Fall.

1. Let p be a prime. Find the number of subgroups H of $\mathbb{Z}_{p^2} \times \mathbb{Z}_{p^3}$, $|H| = p^2$.
Proof. By the fundamental theorem of finitely generated abelian group,
either $|H| \cong \mathbb{Z}_{p^2}$ or $\mathbb{Z}_p \times \mathbb{Z}_p$.

If $|H| \cong \mathbb{Z}_p \times \mathbb{Z}_p$, then for all $x \in H$, $p \cdot x = 0$. Therefore,
 $H \subseteq \{(a, b) \in \mathbb{Z}_{p^2} \times \mathbb{Z}_{p^3} \mid p \cdot (a, b) = 0\}$.

Meanwhile, $pa = 0$ and $pb = 0$ if and only if $a \in \langle p \rangle \subseteq \mathbb{Z}_{p^2}$ and $b \in \langle p \rangle \subseteq \mathbb{Z}_{p^3}$,
so $\{(a, b) \in \mathbb{Z}_{p^2} \times \mathbb{Z}_{p^3} \mid p \cdot (a, b) = 0\} = \langle p \rangle \times \langle p^2 \rangle$.

Since $\langle p \rangle \times \langle p^2 \rangle$ has already order p^2 , so $H = \langle p \rangle \times \langle p^2 \rangle$.

If $H \cong \mathbb{Z}_{p^2}$, that is, H is generated by order p^2 element,
Since $x \in \mathbb{Z}_{p^2} \times \mathbb{Z}_{p^3}$ has order $p^2 \iff p^2 \cdot x = 0$ and $px \neq 0$.

Every element $a \in \mathbb{Z}_{p^2}$ satisfies $p^2 \cdot a = 0$, and $b \in \mathbb{Z}_{p^3}$ satisfies $p^2 \cdot b = 0$
if and only if $b \in p\mathbb{Z}_{p^3}$ and $\mathbb{Z}_{p^3}$ has number of p^2 of p -multiples,
so there are $p^2 \cdot p^2 = p^4$ elements satisfying $p^2 \cdot x = 0$.

And, $px = 0$ if and only if $a \in p\mathbb{Z}_{p^2}$ and $b \in p^2\mathbb{Z}_{p^3}$, where $x = (a, b)$.
So, there are p^2 elements satisfying $px = 0$.

Consequently, there are $p^4 - p^2$ elements x such that $|H| = \langle x \rangle$.

Moreover, the cyclic group of order p^2 has $\phi(p^2) = p^2 - p$ generators,

so the possible $|H| \cong \mathbb{Z}_{p^2}$ are

$$(p^4 - p^2) / (p^2 - p)$$

numbers.

Finally, the number of $|H|$ is $1 + (p^4 - p^2) / (p^2 - p)$.

Note: for a prime p , $\phi(p^n) = p^n - p^{n-1}$.

2020 Fall.

2. For each $n \in \mathbb{N}$, let $a_n = \cos(2\pi/n)$.

a). Prove that a_n is algebraic over $\mathbb{Q}$.

b). Calculate $\deg_{\mathbb{Q}}(a_5)$.

Proof. a). Let $\zeta = \exp(2\pi i/n)$. Then, $\zeta^{-1} = \exp(-2\pi i/n) = \cos(2\pi/n) - i\sin(2\pi/n)$ gives that $a_n = \cos(2\pi/n) = (\zeta + \zeta^{-1})/2$.

Meanwhile, since $\zeta^n = \exp 2\pi i = 1$, so $\zeta^n - 1$ gives ζ is algebraic over $\mathbb{Q}$.

Therefore, ζ^{-1} is also algebraic, thus $a_n = \frac{\zeta + \zeta^{-1}}{2}$ is algebraic over $\mathbb{Q}$.

b). Let $w = \exp(2\pi i/5)$. Since w is the primitive 5th root of unity, $w^4 + w^3 + w^2 + w + 1 = 0$. And, dividing by w^2 then $2a_5 = w + w^{-1}$ implies

$$w^4 + w^3 + w^2 + w + 1 = 0 \Rightarrow w^2 + w^{-2} + w + w^{-1} + 1 = 0$$

$$\Rightarrow 4a_5^2 - 2 + 2a_5 + 1 = 0$$

$$\Rightarrow 4a_5^2 + 2a_5 - 1 = 0.$$

Since $4x^2 + 2x - 1$ is irreducible over $\mathbb{Q}$ by the discriminant $2^2 + 16 = 20$ but $\sqrt{20} \notin \mathbb{Q}$, so $\deg_{\mathbb{Q}}(a_5) = 2$. $\square$

Fall.

a) Notice that De Moivre's formula gives

$$e^{i \cdot n z} = (\cos n z + i \cdot \sin n z) = (\cos z + i \cdot \sin z)^n$$

Therefore

$$\begin{aligned} \cos n \cdot \frac{2\pi}{n} = 1 &= \operatorname{Re} \left(\cos \frac{2\pi}{n} + i \cdot \sin \frac{2\pi}{n} \right)^n \\ &= \operatorname{Re} \left(\sum_{k=0}^n \binom{n}{k} (\cos \frac{2\pi}{n})^k \cdot (i \cdot \sin \frac{2\pi}{n})^{n-k} \cdot \binom{n}{k} \right) \end{aligned}$$

Meanwhile, $\sin(\frac{2\pi}{n})^2 = 1 - \cos(\frac{2\pi}{n})^2$ and

the real part of $\sum_{k=0}^n \binom{n}{k} \cdot (\cos \frac{2\pi}{n})^k \cdot (i \cdot \sin \frac{2\pi}{n})^{n-k}$ yield only $n-k$ is even,

since i^{n-k} is real $\Leftrightarrow n-k$, so $\operatorname{Re} \left(\sum_{k=0}^n \binom{n}{k} \cdot (\cos \frac{2\pi}{n})^k \cdot (i \cdot \sin \frac{2\pi}{n})^{n-k} \right) = 0$ is evaluated $\cos(\frac{2\pi}{n})$ for some polynomial over $\mathbb{Q}$, so a_n is algebraic over $\mathbb{Q}$.

b). From above argument,

$$\begin{aligned} 1 &= \underbrace{1 \cdot a_5^0}_{k=0} + \underbrace{5 \cdot a_5^1 \cdot (1 - a_5^2)}_{k=1} - \underbrace{10 \cdot a_5^3 \cdot (1 - a_5^2)^2}_{k=3} + \underbrace{a_5^5}_{k=5} \\ &= 5 \cdot a_5 - 5 \cdot a_5^3 - 10 a_5^3 \cdot (1 - 2a_5^2 + a_5^4) + a_5^5 \\ &= 5 \cdot a_5 - 15 a_5^3 + 11 a_5^5 - 10 \cdot a_5^7. \end{aligned}$$

$$\text{So, } 10 a_5^7 - 11 a_5^5 + 15 a_5^3 - 5 a_5 + 1 = 0.$$

2020, Fall

3. Let R be a field. Show that $(x^3 - y^5) \in R[x, y]$ is prime ideal.

Proof. Since $R[x, y]$ is U.F.D., every irreducible element is prime.

Thus, enough to show that $x^3 - y^5$ is irreducible.

Notice that: $R[x, y] = R[y][x]$ by definition, so $x^3 - y^5 \in R[y][x]$.

Since $\gcd(1, y^5) = 1$, $x^3 - y^5$ is primitive, by the Gauss Lemma,

$x^3 - y^5$ is irreducible over $R[y] \Leftrightarrow x^3 - y^5$ is irreducible over $R(y)$.

But, since $x^3 - y^5$ has degree 3, $x^3 - y^5$ is reducible only if it has a linear factor, in other say, $x^3 - y^5$ has a root in $R(y)$.

Suppose that for some $f(y)/g(y) \in R(y)$,

$$\frac{f(y)^3}{g(y)^3} - y^5 = 0.$$

Then, $f(y)^3 = y^5 g(y)^3$, But $\deg f(y)^3 \equiv 0 \pmod{3}$, and

$$\deg y^5 g(y)^3 = 5 + \deg g(y)^3 \equiv 2 \pmod{3}.$$

Therefore, there is no such a root, so $x^3 - y^5$ is irreducible over $R(y)$,
so it is irreducible over $R[x, y]$.

Note:

$$K[x, y] = K[y][x] \xrightarrow{\text{Gauss}} K(y)[x] \xrightarrow{\text{irr. Test.}}$$

Modified

Prove that $\gcd(m,n)=1$ implies $x^m - y^n \in F[x,y]$ is irreducible.

2020. Fall.

4. Let $W \subseteq V = \mathbb{R}^n$ be a subspace. Show that there exists a homogeneous system of linear equations whose solution space is W .

Proof. Let $\mathcal{B} = \{\alpha_1, \dots, \alpha_k\}$ be a basis for W .

Then, we can extend the basis $\mathcal{B}$ to $\overline{\mathcal{B}} = \mathcal{B} \cup \{\alpha_{k+1}, \dots, \alpha_n\}$ for V .

Now, consider the map:

$$f_{\mathcal{B}}: V \rightarrow \mathbb{R}^k; \sum c_i \alpha_i \mapsto C_{\mathcal{B}}$$

Then, $W = \ker f_{k+1} \cap \dots \cap \ker f_n$, so for the standard basis $\mathcal{S}$ for V ,

$$A = \begin{pmatrix} [f_{k+1}]_{\mathcal{S}} \\ \vdots \\ [f_n]_{\mathcal{S}} \end{pmatrix} \text{ satisfies } \ker A = W, \text{ so}$$

([f] $_{\mathcal{S}}$ is row vector representation).

$$A \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \text{ is the system we desired.}$$

2021. Spring.

1. Show that no group of 2020 is simple.

Proof $2020 = 2^2 \cdot 5 \cdot 101$. Denote n_p as the number of Sylow p -subgroups.

By the Sylow theorem, $n_{101} \equiv 1 \pmod{101}$ and $n_{101} \mid 20$, so n_{101} must be 1. Therefore, the Sylow 101-subgroup is normal so proved.

Without using Sylow theorem.

By the Cauchy's theorem, G has cyclic subgroup $H \leq G$ of order 101.

Denote $A = \{g \mid g^{-1} \in G\}$. Then, A is orbit of H from the action

$$G \times \{S \leq G\} \rightarrow \{S \leq G\} : (g, S) \mapsto gSg^{-1}.$$

Since the stabilizer of this action of H is $G_H = N_G(H)$, we obtain

$$\#A = |G : N_G(H)|.$$

And since $H \leq N_G(H)$,

$$|G : N_G(H)| \leq |G : H| = 2020/101 = 20, \text{ so } \#A \leq 20.$$

Now, consider the action

$$H \times A \rightarrow A : (h, K) \mapsto hKh^{-1}.$$

Given $K \in A$, the H -orbit of this action is $\text{Orb}_H(K) = \{hKh^{-1} \mid h \in H\}$ and

$$\#\text{Orb}_H(K) = |H : \text{Stab}_H(K)|, \text{Stab}_H(K) \leq H \Rightarrow |\text{Stab}_H(K)| \in \{1, 101\}.$$

So, $\#\text{Orb}_H(K) \in \{1, 101\}$ and $\text{Orb}_H(K) \leq A$, so $\#\text{Orb}_H(K) = 1$.

Therefore, every element in H normalizes $K \in A$, that is, $H \leq N_G(K)$.

Now, for some $K \in A$, if $H \neq K$ then $HK \leq G$ and

$$|HK| = \frac{|H| \cdot |K|}{|H \cap K|} = 101^2$$

but, since $101^2 \nmid 2020$, so $A = \{1\}$. That is, $H \leq G$ $\square$

2021. Spring

7. Let f be a polynomial given by

$$f(z) = a_0 + a_1 z + \dots + a_d z^d \quad \text{over } \mathbb{C}, \quad a_d \neq 0$$

8m.

Evaluate the value $\lim_{R \rightarrow \infty} \int_{|z|=R} \frac{z \cdot f'(z)}{f(z)} dz$

Proof. By the fundamental theorem of Algebra, f has d -roots $z_1, \dots, z_d \in \mathbb{C}$.

That is, $f(z) = a_d(z - z_1) \dots (z - z_d)$

Moreover, $\frac{f'(z)}{f(z)} = \frac{1}{z - z_1} + \dots + \frac{1}{z - z_d}$, just calculation.

Therefore, for $R > \max\{|z_1|, \dots, |z_d|\}$, by the residue theorem,

$$\frac{1}{2\pi i} \int_{|z|=R} \frac{z \cdot f'(z)}{f(z)} dz = \sum_{n=1}^d \operatorname{Res}\left(\frac{z}{z - z_n}, z_n\right) = \sum_{n=1}^d \lim_{z \rightarrow z_n} z = z_1 + \dots + z_d$$

Consequently, the value is

$$2\pi i \cdot (z_1 + \dots + z_d) = 2\pi i \cdot \left(-\frac{a_{d-1}}{a_d}\right),$$

□

2022 Spring.

8. Define $f: \mathbb{R} \rightarrow \mathbb{R}; x \mapsto \sum_{n=1}^{\infty} \frac{1}{x^2 + n^2}$

Then, f is differentiable on $\mathbb{R}$.

Proof. For each $m \in \mathbb{N}$, define

$$f_m(x) = \sum_{n=1}^m \frac{1}{x^2 + n^2}.$$

At first, $f_m(x) \rightarrow f(x)$ as $m \rightarrow \infty$, because that

$$\sum_{n=2}^m \frac{1}{n^2} < \sum_{n=2}^m \frac{1}{n^2 - n} = \sum_{n=2}^m \frac{1}{n-1} - \frac{1}{n} = 1 - \frac{1}{m} \rightarrow 1 \text{ as } m \rightarrow \infty,$$

so the increasing bounded sequence $f_m(x)$ converges. Now,

$$f'_m(x) = \sum_{n=1}^m -\frac{2x}{(x^2 + n^2)^2}$$

converges uniformly, because

if $|x| \leq 1$, then

$$\left| \frac{-2x}{(x^2 + n^2)^2} \right| \leq \frac{2}{n^4} \text{ and } \sum \frac{1}{n^4} < \infty,$$

if $|x| > 1$, then

$$\left| \frac{-2x}{(x^2 + n^2)^2} \right| \leq 2 \cdot \frac{|x|}{x^2 + n^2} \cdot \frac{1}{x^2 + n^2} \leq 2 \cdot \frac{1}{n^2} \text{ and } \sum \frac{1}{n^2} < \infty$$

therefore, the Weierstrass M-test gives f'_m converges uniformly.

Now, f_m converges at some point and f'_m converges uniformly on $\mathbb{R}$, the limit function f is differentiable. $\square$ sm.

~~First, for any $x \in \mathbb{R}$, $|x^2 + n^2| \geq n^2$, so~~

~~$$\sum_{n=1}^{\infty} \left| \frac{1}{x^2 + n^2} \right| \leq \sum_{n=1}^{\infty} \frac{1}{n^2} < \infty, \text{ because } \sum_{n=2}^m \frac{1}{n^2} < \sum_{n=2}^m \frac{1}{n^2 - n} = \sum_{n=2}^m \frac{1}{n-1} - \frac{1}{n} = 1 - \frac{1}{m} \rightarrow 1,$$~~

~~as $m \rightarrow \infty$. Therefore, given ε_0 , for some $N \in \mathbb{N}$, $m > n \geq N$ implies~~

~~$$\sup_{x \in \mathbb{R}} \left| \sum_{k=n}^m \frac{1}{x^2 + k^2} \right| \leq \sup_{x \in \mathbb{R}} \sum_{k=n}^m \left| \frac{1}{x^2 + k^2} \right| < \sum_{k=n}^m \frac{1}{k^2} < \varepsilon.$$~~

~~Now, define for each $m \in \mathbb{N}$,~~

~~$$f_m(x) = \sum_{n=1}^m \frac{1}{x^2 + n^2}.$$~~

~~We already shown $f_m \rightarrow f$ uniformly. And,~~

~~$$f'_m(x) = \sum_{n=1}^m -\frac{2x}{(x^2 + n^2)^2}$$~~

~~converges uniformly, because~~

2022. Fall.

1. Let G be a group and H and K are normal subgroup of G .

If $H \cap K = \{e\}$, then $H/K \cong H \times K$.

Proof. Define a map

$$\varphi: H \times K \rightarrow H/K: (h, k) \mapsto hK.$$

It is well-defined by the definition of H/K , and it is homomorphism; given $h_1, h_2 \in H$ and $k_1, k_2 \in K$,

$$\varphi((h_1, k_1) \cdot (h_2, k_2)) = \varphi((h_1 h_2, k_1 k_2)) = h_1 h_2 K_1 k_2$$

meanwhile, given $h \in H$ and $k \in K$, $hkh^{-1}k^{-1} \in K$ and $\in H$, because $hkh^{-1} \in K$ and $kh^{-1}k^{-1} \in H$, by the normality. But, $H \cap K = \{e\}$ implies $hkh^{-1}k^{-1} = e$, so $hk = kh$, these commute so,

$$h_1 h_2 k_1 k_2 = h_1 k_1 h_2 k_2 = \varphi((h_1, k_1)) \varphi((h_2, k_2)), \text{ so } \varphi \text{ preserves the operation.}$$

And, $\varphi[H \times K] = H/K$ by just definition, φ is onto. Now, the first isomorphism theorem gives that $H \times K / \ker \varphi \cong H/K$.

$$\text{And, } \ker \varphi = \{(h, k) \in H \times K \mid hK = e\} = \{e\},$$

so $H \times K \cong H \times K / \{e\}$, it proves the assertion. $\square$

2022 Fall.

8. For continuous functions $g: [0,1] \rightarrow \mathbb{R}$ and $h: [-1,1] \rightarrow \mathbb{R}$, define

$$f: [0,1] \rightarrow \mathbb{R}; x \mapsto \int_0^1 h(x-y)g(y) dy$$

For each $n \in \mathbb{N}$, define

$$f_n: [0,1] \rightarrow \mathbb{R}; x \mapsto \frac{1}{n} \cdot \sum_{i=0}^{n-1} h(x - \frac{i}{n}) \cdot g(\frac{i}{n})$$

Prove that $f_n \rightarrow f$ uniformly on $[0,1]$

Proof. Define $l: [0,1] \times [0,1]; (x,y) \mapsto h(x-y) \cdot g(y)$.

Since h and g both are continuous, therefore l is continuous and since $[0,1]^2$ is compact set, so l is continuous uniformly.

Given $\epsilon > 0$, there exists $\delta > 0$ such that for any $(x_1, y_1), (x_2, y_2) \in [0,1]^2$,

$$\|(x_1, y_1) - (x_2, y_2)\| < \delta \Rightarrow \|l(x_1, y_1) - l(x_2, y_2)\| < \epsilon.$$

Now, given $x \in [0,1]$, and $n > \frac{1}{\delta}$, ($n > \frac{1}{\delta}$ gives $\frac{1}{n} < \delta$)

$$\begin{aligned} |f(x) - f_n(x)| &= \left| \int_0^1 l(x, y) dy - f_n(x) \right| \\ &= \left| \sum_{i=0}^{n-1} \int_{\frac{i}{n}}^{\frac{i+1}{n}} l(x, y) dy - \frac{1}{n} \cdot \sum_{i=0}^{n-1} l(x, \frac{i}{n}) \right| \\ &= \left| \sum_{i=0}^{n-1} \int_{\frac{i}{n}}^{\frac{i+1}{n}} l(x, y) dy - \sum_{i=1}^{n-1} \int_{\frac{i}{n}}^{\frac{i+1}{n}} l(x, \frac{i}{n}) dy \right| \\ &\leq \sum_{i=0}^{n-1} \int_{\frac{i}{n}}^{\frac{i+1}{n}} |l(x, y) - l(x, \frac{i}{n})| dy \\ &< \sum_{i=0}^{n-1} \int_{\frac{i}{n}}^{\frac{i+1}{n}} \epsilon dy = \epsilon. \end{aligned}$$

Therefore, $n > \frac{1}{\delta} \Rightarrow \sup_{x \in [0,1]} |f_n(x) - f(x)| < \epsilon$, So, proved. $\square$

12022. Fall.

18. Let S be the subset of $\mathbb{R}^3$ given by $\{(x, y, z) \mid z^2 = x^2 + y^2, z \geq 0\}$.

Show that S is not regular surface.

Proof. Suppose that S is a regular surface. Then, for $(0, 0, 0) \in S$, there exists a neighbor V containing $(0, 0, 0)$ such that $V \cap S$ is a graph of

$$z = f(x, y) \quad \text{or} \quad y = g(x, z) \quad \text{or} \quad z = h(y, x)$$

where f, g, h are differentiable. Since projection of S into xz , yz -plane are not 1-1, g, h cannot be such a map.

Meanwhile, $f(x, y) = \sqrt{x^2 + y^2}$ at locally $(0, 0, 0)$, but f is not differentiable at $(x, y) = (0, 0)$, so it cannot be possible. Therefore S is not regular.

(More specifically, $(0, \pm \epsilon, \sqrt{1 - \epsilon^2}) \in S$ for any $\epsilon > 0$, so there is no 1-1 yz -projection, similarly xz .)

2023. Spring

4. Let $V = \{\text{id}, (12)(34), (13)(24), (14)(23)\} \subseteq S_4$, 57,

a). Show that V is normal subgroup.

b). $S_4/V \cong S_3$,

a). Since conjugate cannot change the order of elements, and cannot change the parity: since the sign function is homomorphism,

$$\text{sgn}(\sigma\tau\sigma^{-1}) = \text{sgn}(\sigma) \cdot \text{sgn}(\sigma^{-1}) \cdot \text{sgn}(\tau) = \text{sgn}(\tau), \text{ for any } \sigma, \tau \in S_n.$$

Now, the fact that V contains all order two even permutations in S_4 , so every conjugate of V is contained in V , so normal. $\downarrow$ 5m

b). Consider S_3 is a subgroup of S_4 , the number 4 fixed. Then,

$$\varphi: S_3 \rightarrow S_4/V: \sigma \mapsto \sigma V$$

is well-defined, and given $\sigma_1, \sigma_2 \in S_3 \leq S_4$,

$$\varphi(\sigma_1\sigma_2) = \sigma_1\sigma_2 V = \sigma_1 V \sigma_2 V = \varphi(\sigma_1) \varphi(\sigma_2)$$

so it's homomorphism. And, since all elements in S_3 fix 4, $S_3 \cap V = \{\text{id}\}$, so $\ker \varphi = \{\text{id}\}$. Now, since $|S_3| = 6 = |\ker \varphi| \cdot |\text{Im } \varphi|$, $|\text{Im } \varphi| = 6$, so $\text{Im } \varphi = S_4/V$, since $|S_4/V| = 6$, so φ is onto.

Now, the first isomorphism theorem gives $S_3 \cong S_4/V$. $\square$

2023 Spring

5. Let $I = (2, x^2+x+1) \subseteq \mathbb{Z}[x]$ be an ideal

show that I is maximal ideal and give a multiplication table for $\mathbb{Z}[x]/I$.

Proof. Define a map

$$\pi: \mathbb{Z}[x] \rightarrow \mathbb{Z}/2\mathbb{Z}[x]; \sum_{n=0}^{\infty} a_n x^n \mapsto \sum_{n=0}^{\infty} (a_n \bmod 2) x^n$$

Then, π is an onto Homomorphism, ($\pi(0) = 0$, $\pi(1) = 1$) with

$\ker \pi = 2\mathbb{Z}[x] = (2)$, so $\mathbb{Z}[x]/\ker \pi = \mathbb{Z}[x]/2\mathbb{Z}[x] \cong (\mathbb{Z}/2\mathbb{Z})[x]$.

Meanwhile, x^2+x+1 is irreducible over $\mathbb{Z}/2\mathbb{Z}$, because it is of degree 2 and $0^2+0+1=1 \neq 0$, $1^2+1+1=1 \neq 0$ in $\mathbb{Z}/2\mathbb{Z}$.

Now, Consider the natural Homomorphism

$$\phi: \mathbb{Z}/2\mathbb{Z}[x] \rightarrow (\mathbb{Z}/2\mathbb{Z}[x])/(x^2+x+1); p(x) \mapsto p(x) + (x^2+x+1)$$

It is trivially onto, so $\phi \circ \pi$ is onto Homomorphism.

Moreover,

$$\ker(\phi \circ \pi) = (\phi \circ \pi)^{-1}[\{0\}] = \pi^{-1}[\phi^{-1}[\{0\}]] = \pi^{-1}[\ker \phi] = \pi^{-1}[(x^2+x+1)]$$

Now,

$$f \in \pi^{-1}[(x^2+x+1)] \Leftrightarrow \pi(f) \in (x^2+x+1)$$

$$\Leftrightarrow \pi(f) = (x^2+x+1) \cdot \pi(g) \text{ for some } g(x) \in \mathbb{Z}[x]$$

$$\Leftrightarrow \pi(f - (x^2+x+1)g) = 0$$

$$\Leftrightarrow f - (x^2+x+1)g \in (2)$$

$$\Leftrightarrow f \in (2) + (x^2+x+1) = (2, x^2+x+1).$$

So, $\mathbb{Z}[x]/\ker(\phi \circ \pi) = \mathbb{Z}[x]/(2, x^2+x+1) \cong (\mathbb{Z}/2\mathbb{Z})[x]/(x^2+x+1)$.

And, since $\mathbb{Z}/2\mathbb{Z}$ is field and x^2+x+1 is irreducible over $\mathbb{Z}/2\mathbb{Z}$, so $(\mathbb{Z}/2\mathbb{Z})[x]/(x^2+x+1)$ is a field, so, the quotient is field if and only if (x^2+x+1) is maximal, so proved.

Since $\mathbb{Z}[x]/I \cong (\mathbb{Z}/2\mathbb{Z})[x]/(x^2+x+1)$, Let $\alpha = x + (x^2+x+1)$.

Then $\alpha^2 = x^2 + (x^2+x+1) = x+1 + (x^2+x+1) = \alpha + 1$, so

	1	α
1	1	α
α	α	$\alpha+1$

is the table,

Generalized in the next page.

Let $I = (a, f) \subseteq R[x]$ be an ideal generated by $a, f \in R[x]$.

Consider the diagram:

$$\begin{array}{ccccc} R[x] & \xrightarrow{\mathcal{Q}} & (R/aR)[x] & \xrightarrow{\pi_f} & (R/aR)[x]/(\bar{f}) \\ & \searrow \pi & \cong \nearrow & & \\ & & R[x]/aR[x] & & \end{array}$$

$$\begin{array}{ll} R[x] \rightarrow R[x]/aR[x] & R[x]/aR[x] \rightarrow (R/aR)[x] \\ p(x) \mapsto p(x) + aR[x] & p(x) + aR[x] \mapsto \overline{p(x)}, \text{ where } \overline{\sum a_n \cdot x^n} = \sum (a_n + aR) \cdot x^n. \end{array}$$

Onto Homomorphism. Isomorphism.

$\Rightarrow \mathcal{Q}: R[x] \rightarrow (R/aR)[x]$ is Onto Homomorphism, so is $\pi_f \circ \mathcal{Q}$ onto Homomorphism.

Moreover,

$$\begin{aligned} \ker(\pi_f \circ \mathcal{Q}) &= (\pi_f \circ \mathcal{Q})^{-1}[\{0\}] = \mathcal{Q}^{-1}[\ker \pi_f] \\ &= \{p(x) \in R[x] \mid \overline{p(x)} \in \ker \pi_f\} = \{p(x) \in R[x] \mid \overline{p(x)} \in (\overline{f_{00}})\} \\ &\stackrel{(*)}{=} \{p(x) \in aR[x] + f_{00}R[x]\} = (a) + (f) = (a, f) = I. \end{aligned}$$

Assume $a \in R$ and $(a) \subseteq R$ is maximal, and $\bar{f} \in (R/aR)[x]$ is irreducible.

Then, R/aR is a field, and therefore $(R/aR)[x]$ is P.I.D.

So, $(\bar{f})$ is maximal, it implies that $(R/aR)[x]/(\bar{f})$ is a field.

Now, by the first-isomorphism theorem,

$$R[x]/(aR[x] + fR[x]) \cong (R/aR)[x]/\bar{f} \cdot (R/aR)[x]$$

Note: (*) is obtained by: Given $p(x) \in R[x]$,

$$\overline{p(x)} \in (\overline{f_{00}}) \Leftrightarrow \overline{p(x)} = \overline{g(x) \cdot f_{00}} \text{ for some } g(x) \in R[x]$$

$$\Leftrightarrow p(x) - g(x)f_{00} \in aR[x] \text{ for some } g(x) \in R[x]$$

Another Solution.

Observe that $\mathbb{Z}[x]/(2, x^2+x+1)$.

Given $f(x) \in \mathbb{Z}[x]$, since the leading coefficient of x^2+x+1 is a unit, so we can apply the division Algorithm.

$$f(x) = g(x) \cdot (x^2+x+1) + r(x) \text{ for some } g, r \in \mathbb{Z}[x] \text{ with } \deg r \leq 1.$$

Therefore, $\mathbb{Z}[x]/(2, x^2+x+1) = \{ax+b + (2, x^2+x+1) \mid a, b \in \mathbb{Z}\}$

And, for any $a, b \in \mathbb{Z}$, $ax+b = (a \bmod 2) \cdot x + (b \bmod 2) + 2nx + 2m$ for some $n, m \in \mathbb{Z}$, so

$$= \{ax+b + (2, x^2+x+1) \mid a, b \in \{0, 1\}\}$$

And, $1 \notin (2, x^2+x+1)$ and $x \notin (2, x^2+x+1)$ and $x+1 \notin (2, x^2+x+1)$ implies these are distinct with zero, and also these are mutually distinct, just simple calculation. Let $\alpha = x+1$.

Now, $\mathbb{Z}[x]/I$ has four elements, $\{0, 1, \alpha, \alpha+1\}$ where $\alpha^2+\alpha+1=0$, so the table

	0	1	α	$\alpha+1$
0	0	0	0	0
1	0	1	α	$\alpha+1$
α	0	α	$\alpha+1$	1
$\alpha+1$	0	$\alpha+1$	1	α

is obtained. Since every non-zero element has an inverse, it is a field.

2023, Spring.

7. Let $f: \mathbb{C} \rightarrow \mathbb{C}$ be an entire satisfying 10m

$$U^2(z) - 2 \cdot V^2(z) = 1 \text{ for all } z \in \mathbb{C}$$

Prove that f is constant.

Proof. Observe that ~~differentiating gives~~ that

$$2U U_x - 2V V_x = 0 \Rightarrow \cancel{U U_x} = V V_x = -V \cdot U_y$$

Observe that: $f(z) = 0 \Leftrightarrow U(z) = V(z) = 0$

But, $U^2 - 2V^2 = 1$, so $f(z) \neq 0$ for all $z \in \mathbb{C}$. Now,

$$\frac{1}{f(z)} = \frac{1}{U^2(z) + V^2(z)} = \frac{1}{1 + 3V^2(z)} \leq 1,$$

therefore by the Liouville's theorem, the bounded analytic function $\frac{1}{f^2}$ is constant, so f is also constant function. $\square$

$$\text{Or, } |f^2| = U^2 + V^2 = 1 + 3V^2 \Rightarrow \frac{1}{f} \leq \frac{1}{\sqrt{1+3V^2}} \leq 1.$$

2023 Spring

10. Let $f: [0, \infty) \rightarrow [0, \infty)$ be a continuous function satisfying

$$\int_0^{\infty} f(x) dx < \infty.$$

a). Prove or disprove that $\lim_{x \rightarrow \infty} f(x) = 0$.

b). If f is continuous uniformly then $\lim_{x \rightarrow \infty} f(x) = 0$? Prove or disprove.
Proof.

a). Define a map on $[0, \infty)$

$$f_n(x) = \begin{cases} 0 & \text{otherwise} \\ n^2(x-n) & \text{if } x \in [n, n + \frac{1}{n^2}] \\ n^2(n + \frac{2}{n^2} - x) & \text{if } x \in [n + \frac{1}{n^2}, n + \frac{2}{n^2}] \end{cases}$$

Then $\int_0^{\infty} f_n = \frac{1}{n}$ and f_n is continuous, for each $n \in \mathbb{N}$.

Let $f = f_1 + f_2 + \dots + f_n + \dots$. Then f is continuous and

$$\int_0^{\infty} f = \sum_{n=1}^{\infty} \frac{1}{n} < \infty.$$

(The continuity is obtained just investigate each $[n, n + \frac{2}{n^2}]$)

But, for $\epsilon = \frac{1}{2}$, for any $\delta > 0$, there is $x > \delta$ such that $|f(x) - 0| = 1 \geq \frac{1}{2}$.
So $\lim_{x \rightarrow \infty} f(x) \neq 0$.

b). Suppose $\lim_{x \rightarrow \infty} f(x) = 0$ is false but f is continuous uniformly.

Let $\epsilon = 2$. Then, for any $n \in \mathbb{N}$, there is $x_n > n$ such that $f(x_n) \geq 2$.

Since f is uniformly continuous, take $\delta > 0$ such that $|x - y| < \delta$ implies $|f(x) - f(y)| < 1$. Then, for each $n \in \mathbb{N}$,

$$\int_{x_n}^{x_n + \delta} f(x) dx \geq \delta \cdot 1$$

So, $\int_0^{\infty} f \geq \sum_{n=1}^N \int_{x_n}^{x_n + \delta} f \geq \delta \cdot N$ diverges, So contradicts.

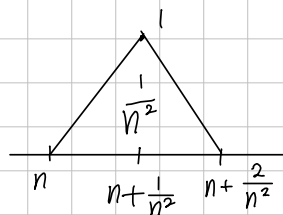
Thus $\lim_{x \rightarrow \infty} f = 0$. $\square$

Revised in the Next.

Let $f: [0, \infty) \rightarrow \mathbb{R}$ be non-negative valued continuous function, does not Converges to Zero, but the 'improper integral' Converges.

Find and Convince Concretely, and change the Condition as f Continuous Uniformly.

Define $f_n(x) = \begin{cases} n^2(x-n) & \text{if } x \in [n, n + \frac{1}{n^2}] \\ n^2(n + \frac{2}{n^2} - x) & \text{if } x \in [n + \frac{1}{n^2}, n + \frac{2}{n^2}] \\ 0 & \text{otherwise.} \end{cases}$



Then, f_n is continuous, with $\int_0^\infty f_n = \frac{1}{n^2}$.

Now, let $f = \sum_{n=2}^\infty f_n$. Since the interval $[n, n + \frac{2}{n^2}]$ are disjoint mutually, $n \geq 2$, so the f is well-defined, but it does not uniformly Converges, because

$$\sup_{x \geq 0} \left| f(x) - \sum_{n=2}^N f_n(x) \right| = 1 \text{ for any } N \in \mathbb{N}.$$

Nevertheless, $\int_0^\infty f = \int_0^\infty \sum_{n=2}^\infty f_n = \sum_{n=2}^\infty \int_0^\infty f_n = \sum_{n=2}^\infty \frac{1}{n^2}$, the integral Converges.

But, for each $n \in \mathbb{N}$, $f(n + \frac{1}{n^2}) = 1$, so $f(x)$ does not tends to Zero as $x \rightarrow \infty$.

Assume that $f(x)$ is Continuous Uniformly

If f does not Converges to Zero, then there is $\varepsilon > 0$ such that for each $R > 0$, there is $x \geq R$ such that $f(x) \geq \varepsilon$.

Now, let $\delta > 0$ such that $|x - y| < \delta \Rightarrow |f(x) - f(y)| < \frac{\varepsilon}{2}$.

And, let $\{x_n\}$ be a Positive Sequence such that for all $n \in \mathbb{N}$,

$$x_{n+1} > x_n + 2\delta \text{ and } f(x_n) \geq \varepsilon.$$

$$\text{Then, } \int_0^{x_n + \delta} f \geq \sum_{k=2}^n \frac{\varepsilon}{2} \cdot 2\delta = n\varepsilon\delta \rightarrow \infty \text{ as } n \rightarrow \infty,$$

so it Contradicts to $\int_0^\infty f < \infty$. $\square$

2024, Spring.

1. Let $A, B \in M_{m \times n}(F)$. Show that $\text{rank}(A+B) \leq \text{rank } A + \text{rank } B$.

[5m]

Proof. By definition,

$$\text{rank}(A+B) = \dim \text{Im}(A+B) \text{ and } \text{rank } A = \dim \text{Im } A, \text{ rank } B = \dim \text{Im } B.$$

Given $\alpha \in \text{Im}(A+B)$, $\alpha = (A+B)\beta = A\beta + B\beta \in \text{Im } A + \text{Im } B$ for some $\beta \in F^n$, therefore $\text{Im}(A+B) \subseteq \text{Im } A + \text{Im } B$. Now,

$$\text{rank}(A+B) \leq \dim(\text{Im } A + \text{Im } B) \leq \dim \text{Im } A + \dim \text{Im } B = \text{rank } A + \text{rank } B.$$

2. Suppose that P and Q are positive definite such that $P^2 = Q^2$. Show $P = Q$.

Proof. Since P is Hermitian, there exists an orthonormal basis $\mathcal{B} = \{\alpha_1, \dots, \alpha_n\}$ such that $P\alpha_i = C_i\alpha_i$ for some $C_i > 0$, for each i .

Let $1 \leq i \leq n$ be fixed. Then, by the assumption,

[5m]

$$P^2\alpha_i = C_i^2\alpha_i = Q^2\alpha_i \Rightarrow (Q^2 - C_i^2 I)\alpha_i = (Q - C_i I)(Q + C_i I)\alpha_i = 0.$$

Since Q is positive definite, $(Q - C_i I)\alpha_i = 0$, because: if it is not zero, then $(Q - C_i I)(Q + C_i I)\alpha_i = (Q + C_i I)(Q - C_i I)\alpha_i = 0$,

so $-C_i$ is negative eigenvalue of Q , it contradicts to Q is positive definite. Therefore, for each $1 \leq i \leq n$,

$$P\alpha_i = C_i\alpha_i = Q\alpha_i,$$

and since $\mathcal{B}$ is basis, therefore $P = Q$. $\square$

2024, Spring

3. Let $T: \mathbb{R}^5 \rightarrow \mathbb{R}^5; x \mapsto Ax$ where

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

a). There exists T -invariant subspace $V \subseteq \mathbb{R}^5$ s.t. $\dim V = 4$?

b). "

s.t. $\dim V = 2$, $T|_V$ is diagonalizable?

c). "

$\dim V = 3$, "

Proof.

a). Let $V = \langle e_1, e_2, e_3, e_4 \rangle$. Then,

$$Ae_1 = e_1, Ae_2 = e_1 + e_2, Ae_3 = e_3, Ae_4 = e_3 + e_4$$

are all contained in V , so $T[V] \subseteq V$. (clearly $\dim V = 4$)

b). Let $V = \langle e_1, e_3 \rangle$. Then $Ae_1 = e_1$, $Ae_3 = e_3$ are contained in V , and

$$[T|_V]_{\{e_1, e_3\}} = I_2. \text{ So } T|_V \text{ is diagonalizable.}$$

c). Observe characteristic polynomial of T is $(t-1)^5$.

If such a subspace V exists, then the characteristic polynomial of $T|_V$ is

$(t-1)^3$. Since $T|_V$ is diagonalizable if and only the minimal polynomial

splits and separable, thus the minimal polynomial of $T|_V$ must be $(t-1)$.

That is, $T|_V = I_V$ but it implies that $\ker(T-I)$ has dimension more than or equal 3, it contradicts to

$$\dim \ker(A-I) = \dim \ker \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} = 2.$$

2024, Spring.

5. Let K be a field, and $L_1, L_2 \subseteq \overline{K}$ be finite extensions of K . 41

a). Prove $L = L_1 L_2$ is finite extension.

b). If L_1 and L_2 are Galois over K , then L/K is Galois.

a). Since L_1 and L_2 both are finite, denote

$$L_1 = K(\alpha_1, \dots, \alpha_n) \quad \text{and} \quad L_2 = K(\beta_1, \dots, \beta_r) \quad \text{where } \alpha_i, \beta_j \in \overline{K} \text{ are algebraic over } K.$$

$$\text{Now, } L = L_1 L_2 = \bigcap_{\substack{E \subseteq \overline{K} \\ L_1, L_2 \subseteq E}} E = K(\alpha_1, \dots, \alpha_n, \beta_1, \dots, \beta_r)$$

is finitely generated by algebraic elements α_i, β_j , so L is finite. 14m.

b). Since L_1 and L_2 are splitting field of K , let f_1 and f_2 be polynomials over K such that L_i is the splitting field of f_i , $i=1,2$. It is possible, because L_1, L_2 both are finite extension.

Now let E be the splitting field of f_1, f_2 . Then, $E \subseteq L_1 L_2$, because E is the smallest field containing all roots of f_1, f_2 , $L_1 L_2 \subseteq E$ because E contains all roots of f_1 and f_2 . So, $L = E$. Therefore L is splitting field over K . 17m.

Since L_1 and L_2 both are Galois, for some separable f_1 and f_2 over K , L_1 and L_2 are splitting field of f_1 and f_2 respectively.

$L_1 L_2$ is f_1, f_2 splitting field $\Rightarrow OK$

즉 f_1, f_2 is separable 이란 보장 X.

중복 linear factor 소거 $\Rightarrow f_1, f_2$ 가 irreducible over K 인가?

Revised in the Algebra/Field Theory/

2024 Spring.

7. Let $f: (-1, 1) \rightarrow \mathbb{R}$ be a differentiable function satisfying $f(0) = 0$ and

$$\text{let } g: (-1, 1) \rightarrow \mathbb{R}: \begin{cases} f(x)/x & \text{if } x \neq 0 \\ f'(0) & \text{if } x = 0. \end{cases}$$

a). If $f \in C^2$, then $g \in C^1$ (prove)

b). If $f \in C^1$ but $\notin C^2$, then $g \in C^1$ may not hold.

Proof. Notice that g is continuous, since

$$\lim_{x \rightarrow 0} g(x) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = f'(0).$$

a) For $x \neq 0$,

$$g'(x) = \left(\frac{f(x)}{x} \right)' = \frac{x f'(x) - f(x)}{x^2} \quad \text{exists, By the L'Hospital Theorem,}$$

$$\lim_{x \rightarrow 0} g'(x) = \lim_{x \rightarrow 0} \frac{f'(0) + x f''(x) - f'(0)}{2x} = \frac{1}{2} \lim_{x \rightarrow 0} f''(x) = \frac{1}{2} f''(0).$$

(Since $x^2 \rightarrow 0$ and $x f'(x) - f(x) \rightarrow 0$ as $x \rightarrow 0$, we can use the theorem).

$$\text{And, } g'(0) = \lim_{x \rightarrow 0} \frac{g(x) - g(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{f(x)/x - f'(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{f(x) - x f'(0)}{x^2} = \frac{1}{2} f''(0).$$

Since the L'Hopital theorem again, so $g \in C^1$.

1/22/24 spring.

8. Let $C_b(\mathbb{R})$ be the set of all real-valued bounded continuous functions on $\mathbb{R}$. For two functions $f, g \in C_b(\mathbb{R})$, define

$$d(f, g) := \sup_{x \in \mathbb{R}} |f(x) - g(x)|.$$

Let $\{f_n\}_{n=1}^{\infty}$ be a sequence in $C_b(\mathbb{R})$ such that:

$\forall \varepsilon > 0, \exists N \in \mathbb{N}$ such that $m, n \geq N \Rightarrow d(f_m, f_n) < \varepsilon$.

a). Prove that there exists $f \in C_b(\mathbb{R})$ such that $\lim_{n \rightarrow \infty} d(f, f_n) = 0$

b). Prove that $\lim_{n \rightarrow \infty} \left(\sup_{x \in \mathbb{R}} |f_n(x)| \right) = \sup_{x \in \mathbb{R}} |f(x)|$.

[Proof. For each $x \in \mathbb{R}$, a sequence $\{f_n(x)\}$ in $\mathbb{R}$ is Cauchy sequence by hypothesis. Since $\mathbb{R}$ is complete, $\lim_{n \rightarrow \infty} f_n(x)$ exists, put this $f(x) \in \mathbb{R}$.

a).

Since $x \in \mathbb{R}$ was arbitrary, $f(x)$ is well-defined on $\mathbb{R}$, but still not necessarily $f \in C_b(\mathbb{R})$, and $d(f, f_n) \rightarrow 0$.

Given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $m > n \geq N$ implies that $\sup_{x \in \mathbb{R}} |f_m(x) - f_n(x)| \leq \varepsilon$.

So, for any $x \in \mathbb{R}$, $|f_m(x) - f_n(x)| \leq \varepsilon$. Letting $m \rightarrow \infty$, we obtain $|f(x) - f_n(x)| \leq \varepsilon$.

Since x was arbitrary, $\sup_{x \in \mathbb{R}} |f(x) - f_n(x)| \leq \varepsilon$, for $n \geq N$.

Therefore, the triangle inequality gives $|f(x)| \leq |f_n(x)| + \varepsilon$ for all $x \in \mathbb{R}$

and since f_n was bounded, f is bounded. And, the argument above

proves $\sup_{x \in \mathbb{R}} |f - f_n| \rightarrow 0$ as $n \rightarrow \infty$, so since f_n is continuous and

$f_n \rightarrow f$ uniformly, so f is also continuous, so $f \in C_b(\mathbb{R})$ and a) is proved.

b). Since $\left| \sup_{x \in \mathbb{R}} |f_n(x)| - \sup_{x \in \mathbb{R}} |f(x)| \right| \leq \sup_{x \in \mathbb{R}} |f_n(x) - f(x)| \rightarrow 0$ as $n \rightarrow \infty$

Prove directly.

2024, Spring.

14. Let X be a compact Hausdorff space. Prove that

X is metrizable if and only if X has a countable basis.

Proof Suppose that X is metrizable by the metric $d: X \times X \rightarrow \mathbb{R}$.

Given $n \in \mathbb{N}$, $\{B_{\frac{1}{n}}(x) \mid x \in X\}$ is an open cover of X , so the compactness gives that $\mathcal{B}_n = \{B_{\frac{1}{n}}(x_i) \mid i \in F_n\}$, the finite subcover exists

with finite index set F_n . Now, put $\mathcal{B} = \bigcup_{n \in \mathbb{N}} \mathcal{B}_n$.

Let $x \in X$ be given and let $B_r(y)$ be an open ball containing x . Then, for $\varepsilon = r - d(x, y)$, $x \in B_\varepsilon(y) \subseteq B_r(y)$, because

$$a \in B_\varepsilon(x) \Leftrightarrow d(a, x) < \varepsilon = r - d(x, y) \Rightarrow d(a, y) = d(a, x) + d(x, y) < r.$$

Now, let $N \in \mathbb{N}$ such that $\frac{1}{N} < \frac{\varepsilon}{2}$. Then, since each $\mathcal{B}_n$ covers X , for some x_i , $x \in B_{\frac{1}{N}}(x_i)$, and

$$z \in B_{\frac{1}{N}}(x_i) \Leftrightarrow d(z, x_i) < \frac{1}{N} \Rightarrow d(z, x) \leq d(z, x_i) + d(x_i, x) < \frac{1}{N} + \frac{1}{N} < \varepsilon.$$

So, $x \in B_{\frac{1}{N}}(x_i) \subseteq B_\varepsilon(x)$, it proves $\mathcal{B}$ is a basis for (X, d) .

and since $\mathcal{B}$ is countable union of finite sets, so it is countable.

Conversely, since compact Hausdorff space is normal, so it is regular. By the Urysohn metrization theorem, second countable regular space is metrizable, so proved.

2024, Fall.

1. Let V be a finite-dimensional vector space over a field F .
Let $T: V \rightarrow V$ be a linear operator such that $T^2 = T$. Prove

η_m

a). $V = \ker T \oplus \operatorname{Im} T$

b). T is diagonalizable.

Proof. a). Given $\alpha \in V$, $\alpha = \alpha - T(\alpha) + T(\alpha)$, $T(\alpha) \in \operatorname{Im} T$ is trivial,
and $T(\alpha - T(\alpha)) = T(\alpha) - T^2(\alpha) = (T - T^2)(\alpha) = 0$, so $\alpha - T(\alpha) \in \ker T$.

Since α was arbitrary, $V = \ker T + \operatorname{Im} T$.

Moreover, let $\alpha \in \ker T \cap \operatorname{Im} T$. Then, $T(\alpha) = 0$ and $\alpha = T(\beta)$ for some $\beta \in V$. Now, $\alpha = T(\beta) = T^2(\beta) = T(T(\beta)) = T(\alpha) = 0$, so $\ker T \cap \operatorname{Im} T = \{0\}$,
it proves $V = \ker T \oplus \operatorname{Im} T$.

b). Since $T^2 - T = 0$, the minimal polynomial m of T divides $x^2 - x$.
If $m = x$ or $x - 1$, then $T = 0$ or $T = \operatorname{Id}$, already diagonal.
If $m = x(x-1)$, then it splits over the given field and separable,
so T is diagonalizable. $\square$

Note: Naturally, $T|_{\operatorname{Im} T} = \operatorname{Id}$, so a) implies b).

2024 Fall.

8. Solve the followings: $x \in \mathbb{R}$,

a). $x''(t) - 2x'(t) + x(t) = 0$

b). $x''(t) - 2x'(t) + x(t) = te^t + 4$.

Proof.

a). Since the characteristic polynomial of the equation is

$$r^2 - 2r + 1 = (r-1)^2,$$

the general solution is

$$x_1(t) = C_1 e^t + C_2 \cdot t e^t \quad \text{where } C_1, C_2 \in \mathbb{R}.$$

b). Calculate the Wronskian:

$$W(e^t, te^t) = \begin{vmatrix} e^t & te^t \\ e^t & e^t + te^t \end{vmatrix} = e^{2t} + te^{2t} - te^{2t} = e^{2t}.$$

Therefore, the particular solution is

$$\begin{aligned} Y(t) &= \int_{t_0}^t \begin{vmatrix} e^u & te^u \\ ue^u & te^u \end{vmatrix} \cdot b(u) / W(e^u, ue^u) du \\ &= \int_{t_0}^t (e^u te^u - ue^u e^u) \cdot (ue^u + 4) / e^{2u} du \\ &= \int_{t_0}^t (ue^{2u} te^u + 4e^u te^u - u^2 e^{2u} e^u - 4ue^u e^u) / e^{2u} du \\ &= \int_{t_0}^t u \cdot te^u + 4 \cdot te^u \cdot e^{-u} - u^2 \cdot e^u - 4e^u \cdot ue^{-u} du \end{aligned}$$

Now, $x_2 + Y$ is the general solution.

2024, Fall,

1. For a fixed $\xi \in \mathbb{R}$, let

$$f(z) = \frac{e^{-2\pi i z \xi}}{\cosh(\pi z)} \quad (z \in \mathbb{C}).$$

a). Show that $\alpha = i/2$ and $\beta = 3i/2$ are simple poles of f .

b). Compute the residues of f at α and β .

c). Evaluate $\int_{-\infty}^{\infty} \frac{e^{-2\pi i x \xi}}{\cosh(\pi x)} dx$.

Proof. a). Notice that $e^{-2\pi i z \xi}$ is entire and has no zero, we need to investigate pole of $1/\cosh(\pi z)$.

Since $\cosh(\pi z) = 0$ if $z_0 = \alpha$ or β , and

$$\lim_{z \rightarrow z_0} \frac{\cosh(\pi z) - \cosh(\pi z_0)}{z - z_0} = \sinh(\pi z_0) \cdot \pi \neq 0, \quad \text{so}$$

$$\lim_{z \rightarrow z_0} (z - z_0) \cdot \frac{1}{\cosh(\pi z)} = \frac{1}{\pi \sinh(\pi z_0)} \text{ exists, so } \alpha \text{ and } \beta \text{ are simple poles.}$$

b). Since α and β both are simple pole,

$$\text{Res}(f, \alpha) = \lim_{z \rightarrow \alpha} (z - \alpha) \cdot f(z) = e^{-2\pi i \alpha \xi} \cdot \frac{1}{\pi \cdot \sinh(\pi \alpha)} = \frac{e^{\pi \xi}}{\pi \cdot \sinh(\pi i/2)} = \frac{e^{\pi \xi}}{\pi i}.$$

$$\text{Res}(f, \beta) = e^{-2\pi i \beta \xi} \cdot \frac{1}{\pi \cdot \sinh(\pi \beta)} = \frac{e^{3\pi \xi}}{\pi \cdot (-i)} = -\frac{e^{3\pi \xi}}{\pi i}.$$

c). Let C_R be the rectangle $C_R = C_1 * C_2 * C_3 * C_4$ along four line segments $-R$ to R , R to $R+i$, $R+i$ to $-R+i$, $-R+i$ to $-R$.

And, observe that

$$f(x+i) = \frac{e^{-2\pi i x \xi + 2\pi \xi}}{\cosh(\pi x + \pi i)} = e^{2\pi \xi} \cdot -f(x)$$

$$\text{Therefore, } \int_{C_1 + C_3} f = \left(\int_{-R}^R f(x) dx \right) \cdot (1 + e^{2\pi \xi}).$$

And, since $|f(R+it)| = \left| \frac{e^{-2\pi i R \xi + 2\pi t \xi}}{\cosh(\pi R + \pi it)} \right| \rightarrow 0$ as $|\cosh(\pi R + \pi it)| \rightarrow \infty$ as $R \rightarrow \infty$.

Therefore, $\int_{C_2} f, \int_{C_4} f \rightarrow 0$, so, by the Residue theorem,

$$\int_{-\infty}^{\infty} f(x) dx = \frac{1}{1 + e^{2\pi \xi}} \cdot \lim_{R \rightarrow \infty} \int_{C_R} f = \frac{1}{1 + e^{2\pi \xi}} \cdot 2e^{\pi \xi}.$$

2025 Spring.

13. Let D^2 be a closed unit disc in $\mathbb{R}^2$ centered at the origin.

a). Show that the quotient space $\mathbb{R}^2/D^2$ is homeomorphic to $\mathbb{R}^2$.

b). Let $U = \text{int } D$. Determine $\mathbb{R}^2/U \cong \mathbb{R}^2$.

Proof. Let $\sim$ be a relation defined on $\mathbb{R}^2$ as:

$$(x_1, y_1) \sim (x_2, y_2) \Leftrightarrow \{(x_1, y_1), (x_2, y_2)\} \subseteq D^2.$$

Then, it is clearly an equivalence relation, and $\mathbb{R}^2/\sim = \mathbb{R}^2/D^2$, by definition.

a). Define a map

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}^2; (x, y) \mapsto$$

Then, given $(x_1, y_1), (x_2, y_2) \in \mathbb{R}^2$,

$$(x_1, y_1) \sim (x_2, y_2) \Leftrightarrow$$

2025. Spring.

14. For $a, b, c > 0$, Consider the ellipsoid

$$E_{a,b,c} = \left\{ (x, y, z) \in \mathbb{R}^3 \mid \left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 + \left(\frac{z}{c}\right)^2 = 1 \right\}.$$

- a). If $a=b$, then $E_{a,a,c}$ is a surface of revolution (Prüferl.)
- b) Give a parametrization of $E_{a,a,c}$. Compute the first fundamental form of this parametrization and the Christoffel symbol.
- c). Show that the Curve $E_{a,b,c} \cap \{x=0\}$ is the image of a geodesic.

a). Let

$$X: (0, 2\pi) \times (0, 2\pi) \rightarrow \mathbb{R}^3; (u, v) \mapsto (a \cdot \cos v \cdot \cos u, a \cdot \cos v \cdot \sin u, c \cdot \sin v).$$

2025, Fall

1. a). Show that $\phi: P_n(\mathbb{R}) \rightarrow \mathbb{R}: p(t) \mapsto p'(0)$ is linear functional.

8m

b). Find a dual basis corresponding to $\{1, t, \dots, t^n\}$ of $P_n(\mathbb{R})$.

Proof. Let $f(x) = \sum_{i=0}^n a_i x^i$ and $g(x) = \sum_{i=0}^n b_i x^i$ and $C \in \mathbb{R}$ be given.

Then, $\phi(Cf + g) = (Cf + g)'|_{t=0} = C \cdot f'|_{t=0} + g'|_{t=0} = C \cdot \phi(f) + \phi(g)$

by the linearity of differentiation. This shows the a),

To show b), let

$$\mathcal{B}^* := \left\{ \frac{1}{k!} \cdot \phi_k: P_n(\mathbb{R}) \rightarrow \mathbb{R}: p(t) \mapsto p^{(k)}(0) \mid k=0, 1, \dots, n \right\}$$

Then, for each $k=0, 1, \dots, n$,

$$\frac{1}{k!} \cdot \phi_k(t^k) = \frac{k!}{k!} \cdot 1 = 1, \text{ and } \frac{1}{k!} \cdot \phi_k(t^r) = 0 \text{ if } r \neq k.$$

Similarly a), the elements in $\mathcal{B}^*$ are all linear functional, so the $\mathcal{B}^*$ is dual basis.

2025. Fall

2. Let $A \in M_{\text{max}}(1\mathbb{R})$.

15m: A^n 의 rank, $n=2$ 일 때부터 시작

a). For $n \in \mathbb{N}$, $\text{rank } A^n \leq \text{rank } A$.

b). Let $L_A: \mathbb{R}^m \rightarrow \mathbb{R}^m, x \mapsto Ax$.

Show that for $n \geq 2$,

$$\text{rank } A^n = \text{rank } A \quad \text{implies} \quad \ker L_A \cap \text{Im } L_A = \{0\}.$$

Sol) a). Given $x \in \text{Im } A^n$, $x = A^n y$ for some $y \in \mathbb{R}^m$.

So, $x = A^n y = A^{n-1} A y \in \text{Im } A$, for $n \geq 2$. Thus $\text{Im } A^n \subseteq \text{Im } A$, it follows $\text{rank } A^n \leq \text{rank } A$. If $n=1$, clear.

b). Given $x \in \ker A$, $A^n x = A^{n-1} A x = A^{n-1} 0 = 0$, so $x \in \ker A^n$, so $\ker A \subseteq \ker A^n$. Since $\text{rank } A^n = \text{rank } A$, the dimension theorem gives $\text{nullity } A^n = \text{nullity } A$.

Therefore, since $\mathbb{R}^m$ is finite dimensional, so is $\ker A^n$, thus $\ker A = \ker A^n$.

Now, let $x \in \ker L_A \cap \text{Im } L_A$ be given. Then, $Ax = 0$ and $x = Ay$ for some $y \in \mathbb{R}^m$.

Then, $A^{n-1} x = A^n y = 0$, because $Ax = 0$, so $A^{n-2} Ax = 0$, and this implies $y \in \ker A^n$.

Since $\ker A^n = \ker A$, $Ay = 0$, so $x = Ay = 0$, so proved. $\square$

2025. Fall

3. Find all $a, b, c \in \mathbb{R}$ such that

6m

$$A = \begin{bmatrix} 2 & 0 & 0 \\ a & 2 & 0 \\ b & c & -1 \end{bmatrix} \text{ is diagonalizable.}$$

Sol. Since the characteristic polynomial of A is regardless of a, b, c ,

$$\chi_A(t) = (t-2)^2(t+1).$$

Since a Matrix is diagonalizable if and only if the minimal polynomial splits and separable, and the minimal polynomial divides the characteristic polynomial, the A is diagonalizable if and only if

$$(A-2I)(A+I) = \begin{bmatrix} 0 & 0 & 0 \\ a & 0 & 0 \\ b & c & -3 \end{bmatrix} \cdot \begin{bmatrix} 3 & 0 & 0 \\ a & 3 & 0 \\ b & c & 0 \end{bmatrix} = 0.$$

$$\text{Calculating gives } = \begin{bmatrix} 0 & 0 & 0 \\ 3a & 0 & 0 \\ ac & 0 & 0 \end{bmatrix}$$

therefore, $a=0$, $b \in \mathbb{R}$, $c \in \mathbb{R}$ are all possible (a, b, c) .

2025. Fall.

4. Let $N \subseteq GL_n(\mathbb{R}) \times GL_n(\mathbb{R})$ be the set of all pairs (A, B) such that $\det A = \pm \det B$.

20m

Show that N is normal and determine the factor group $(G \times G)/N$.

Sol). Let $(A, B) \in N$ and $(C, D) \in GL_n(\mathbb{R}) \times GL_n(\mathbb{R})$ be given. Then

$$(C, D)(A, B)(C, D)^{-1} = (CAC^{-1}, DBD^{-1})$$

And, since the determinant preserves the multiplication of matrix,

$$\det CAC^{-1} = \det C \cdot \det C^{-1} \cdot \det A = \det A$$

$$\text{and } \det DBD^{-1} = \det D \cdot \det D^{-1} \cdot \det B = \det B$$

So, since $(A, B) \in N$, $\det CAC^{-1} = \det A = \pm \det B = \pm \det DBD^{-1}$

therefore $(C, D)(A, B)(C, D)^{-1} \in N$. And, to show N is subgroup,

first, $(I, I) \in N$ trivially, so $N \neq \emptyset$. Let $(A, B), (C, D) \in N$ be given. Then

$$(A, B) \cdot (C, D)^{-1} = (AC^{-1}, BD^{-1})$$

satisfies $\det AC^{-1} = \det A \cdot \det C^{-1} = \pm \det B \cdot \det C^{-1} = \pm \det B \cdot \det D^{-1} = \pm \det BD^{-1}$.

therefore, by the subgroup criterion, N is subgroup, and normal. $\perp$ com.

Define a map

$$\mathcal{Q}: GL_n(\mathbb{R}) \times GL_n(\mathbb{R}) \rightarrow \mathbb{R}^{\times}; (A, B) \mapsto (\det A - \det B)^2$$

Then, $\ker \mathcal{Q} = N$ and $\mathcal{Q}$ is homomorphism, onto $(0, \infty)$, because

the det preserves multiplication, so is $\mathcal{Q}$, and onto is given by: fix $B = I$, and $A = \text{diag}(r, 1, \dots, 1)$ where $r > 0$ is arbitrary.

By the first isomorphism theorem,

$$GL_n(\mathbb{R}) \times GL_n(\mathbb{R}) / N \cong ((0, \infty), \times) \quad \perp \text{ com.} \quad \textcircled{2}$$

Comment: $\mathcal{Q}$ 은 먼저 $\mathbb{R}^{\times}$ 라고 시작하고, hom 임을 보려면 $\mathcal{Q}$ 가 $\times$ 에 대해 호모가 되는 것을 보아야 함!

12-25. Fall

10m

5. Let G be an order 21 subgroup of $\mathbb{C}^\times$.

a) G is cyclic.

b). For any $g \in G$, $x^{12} = g^{15}$ has 3 distinct solutions in G .

Proof. a). Since $\mathbb{C}$ is abelian, so G is also abelian.

By the Cauchy's theorem, G has order 3 element x and order 7 element y .
Now, since 3 and 7 are relatively prime, $|xy| = 21$, so $G = \langle xy \rangle$.

b). By a), $G \cong \mathbb{Z}/21\mathbb{Z}$. So, we can write the given equation as

$$12x \equiv 15g \pmod{21} \quad (g \in \mathbb{Z}/21\mathbb{Z}).$$

Regardless of g , $\gcd(12, 21) = 3$ divides $15g$, so it has three distinct solutions.

Note: $ax \equiv b \pmod{n}$ has a solution iff $\gcd(a, n) \mid b$.

And in such a case, it has $\gcd(a, n)$ solutions.

2025. Fall.

6. Let $\{a_n\}$ be a bounded sequence in $\mathbb{R}$. Define

22m

$$\{a_n\}^* = \{x \in \mathbb{R} \mid x \text{ is a subsequential limit of } \{a_n\}\}$$

a). Prove that $\{a_n\}^*$ is closed.

b). Show $\limsup_{n \rightarrow \infty} a_n = \sup \{a_n\}^*$

c). Is there a sequence $\{b_n\}$ such that $\{b_n\}^* = [0, 1]$?

Proof a). If $\{a_n\}^*$ is finite, then clearly closed in $\mathbb{R}$.

Assume that $\{a_n\}^*$ is infinite. Since $\{a_n\}$ is bounded, so the set of subsequential limits $\{a_n\}^*$ is bounded set. So it has a limit point.

Let α be a limit point of $\{a_n\}^*$. Then, given $\epsilon \in \mathbb{R}$, there exists a

subsequential limit p of $\{a_n\}$ such that $p \in B_{\frac{\epsilon}{2}}(\alpha) \setminus \{\alpha\}$.

For $k=1$, choose such a $p_1 \in B_{\frac{\epsilon}{2}}(\alpha) \setminus \{\alpha\}$, and since p is a subsequential limit, choose $a_{p_1} \in B_{\frac{1}{k}}(p)$, then $a_{p_1} \in B_{\frac{\epsilon}{2}}(\alpha)$.

For chosen $a_{p_1}, \dots, a_{p_{k-1}}$, choose $p_k \in B_{\frac{\epsilon}{2}}(\alpha) \setminus \{\alpha\}$.

Then, similarly, there is $a_{p_k} \in B_{\frac{\epsilon}{2}}(\alpha)$ such that $p_k > p_{k-1}$.

because no such p_k exists, then it is contradiction to a_{p_k} is a subsequential limit. Therefore, the arbitrary limit point α is contained in $\{a_n\}^*$,

so it is closed.

b). Since a), $\{a_n\}^*$ is closed and bounded, so compact. Thus

$$\sup \{a_n\}^* = \max \{a_n\}^* = \alpha \text{ for some subsequential limit } \alpha.$$

So, given $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that $n \geq N \Rightarrow a_n < \alpha + \epsilon$ (since $\alpha = \max \{a_n\}^*$)

and there exists infinitely many terms in $\{a_n\}$ such that $\alpha - \epsilon < a_n$ (since $a_{p_k} \rightarrow \alpha$)

Thus $\limsup_{n \rightarrow \infty} a_n = \alpha$

c). Let b_n be defined as:

$$\frac{1}{1}, \frac{1}{2}, \frac{2}{2}, \frac{1}{3}, \frac{2}{3}, \frac{3}{3}, \dots$$

Then, given $b \in [0, 1]$, there exists a subsequence in $\{b_n\}$ converges to b , because: for any $\epsilon > 0$, there are infinitely many terms in b_n such that

$$|b_n - b| < \epsilon,$$

so we can construct the subsequence. (20)

2025, Fall.

7. Let E be a nonempty bounded subset of a Metric space (X, d) .

10m

Define $\text{diam } E := \sup\{d(x, y) \mid x, y \in E\}$

a). Prove $\text{diam } E = \text{diam } \bar{E}$

b). If E is compact, then $d(x, y) = \text{diam } E$ for some $x, y \in E$.

Proof.

a). Since $E \subseteq \bar{E}$, $\text{diam } E \leq \text{diam } \bar{E}$ is clear.

To show the inverse inequality, using the property of supremum.

Given $\varepsilon > 0$, there exists $x, y \in \bar{E}$ such that

$$\text{diam } \bar{E} - \frac{\varepsilon}{2} < d(x, y) \leq \text{diam } \bar{E}$$

And, since $\bar{E}$ is closure, there are $x', y' \in E$ such that

$$d(x', x) < \frac{\varepsilon}{4} \quad \text{and} \quad d(y, y') < \frac{\varepsilon}{4}.$$

Now, $\text{diam } \bar{E} < d(x, y) + \frac{\varepsilon}{2} < d(x, x') + d(x', y') + d(y', y) + \frac{\varepsilon}{2} < d(x', y') + \varepsilon$.

Since $\varepsilon > 0$ was arbitrary, $\text{diam } \bar{E} \leq \text{diam } E$, so proved. 7m.

b). Since E is compact set, so $E \times E$ is also compact set.

Since the metric function d is continuous, it preserves compactness, so $d[E \times E] \subseteq \mathbb{R}$ is compact set. Therefore,

$$\text{diam } E = \max d[E \times E] \in d[E \times E],$$

so there is $(x', y') \in E \times E$ such that $d(x', y') = \max d[E \times E]$ 3m.

2025 Fall.

8. Let $a > 1$. Evaluate

5m

$$\lim_{t \rightarrow 0^+} \int_t^{at} \frac{\cos x}{x} dx$$

Proof. Consider the Taylor expansion

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$$

Therefore $\frac{\cos x}{x} = \frac{1}{x} - \frac{x}{2!} + \frac{x^3}{4!} - \dots$

Now,
$$\int_t^{at} \frac{\cos x}{x} dx = \int_t^{at} \frac{1}{x} + \int_t^{at} -\frac{x}{2!} + \frac{x^3}{4!} - \dots dx$$
$$= \ln a + \sum_{n=1}^{\infty} \int_t^{at} (-1)^n \frac{x^{2n-1}}{(2n)!} dx$$

(It is possible, because $\sum (-1)^n x^{2n-1}/(2n)!$ converges for all $x \in \mathbb{R}$)

Now,
$$\lim_{t \rightarrow 0^+} \sum_{n=1}^{\infty} \int_t^{at} (-1)^n \frac{x^{2n-1}}{(2n)!} dx = 0,$$
 so

the given limit is $\ln a$.

Revised in next

Observation for $\lim_{t \rightarrow 0^+} \int_t^a \frac{\cos x}{x} dx$.

As we know, the power series representation

$$\frac{1}{x} \cdot \cos x = \frac{1}{x} \cdot \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = \frac{1}{x} + \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n-1}}{(2n)!}$$

is valid for all $x \neq 0$. Therefore, for any $t > 0$,

$$\begin{aligned} \int_t^a \frac{\cos x}{x} dx &= \int_t^a \frac{1}{x} + \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n-1}}{(2n)!} dx \\ &= \int_t^a \frac{1}{x} dx + \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} \Big|_t^a \end{aligned}$$

Meanwhile, the series converges uniformly in $[t, a]$ for small enough $t > 0$, specifically, $t < \frac{1}{a}$, then for each $n \in \mathbb{N}$,

$$\left| (-1)^n \frac{x^{2n}}{(2n)!} \right| \leq \left| \frac{1}{(2n)!} \right| \quad \text{and} \quad \sum_{n=1}^{\infty} \frac{1}{(2n)!} < \infty,$$

therefore the Weierstrass M-test gives the condition,

Thus, we can exchange the integral to the sum,

$$\int_t^a \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n-1}}{(2n)!} = \sum_{n=1}^{\infty} \int_t^a (-1)^n \frac{x^{2n-1}}{(2n)!} = \sum_{n=1}^{\infty} (-1)^n \left(\frac{1}{2n \cdot (2n)!} \cdot (a^{2n} - t^{2n}) \right)$$

Converges uniformly similarly, now, we can exchange the limit to the sum

$$\begin{aligned} &\lim_{t \rightarrow 0^+} \sum_{n=1}^{\infty} (-1)^n \frac{1}{2n \cdot (2n)!} \cdot t^{2n} (a^{2n} - 1) \\ &= \sum_{n=1}^{\infty} \lim_{t \rightarrow 0^+} \quad // \\ &= \sum_{n=1}^{\infty} 0 \\ &= 0 \end{aligned}$$

Consequently, the given integral is

$$\int_t^a \frac{1}{x} dx = \ln a - \ln t = \ln a.$$

Note: $\frac{\cos x}{x} = \frac{1}{x} + \frac{\cos x - 1}{x}$, $\left| \int_t^a \frac{\cos x - 1}{x} dx \right| \leq \int_t^a \frac{x}{2} \rightarrow 0$ as $t \rightarrow 0$.

2025. Fall.

12. Define an equi. rel on $\mathbb{R}$ as $x \sim y \Leftrightarrow x - y \in \mathbb{Q}$.

" as $x \sim y \Leftrightarrow x = y$ or $x - y \in \mathbb{Q}$.

a). Describe the quotient topology on $\mathbb{R}/\sim$.

b). " on $\mathbb{R}/\sim$.

c). In $\mathbb{R}/\sim$, determine if every point is closed.

d). " the image of the set of irrationals under quotient map is compact.

Proof. Since $\mathbb{Q}$ is closed under the operation $-$, so these relations are equi. rel.

a). First, $\mathbb{R}/\sim = \{a + \mathbb{Q} \mid a \in \mathbb{R}\}$, where $a + \mathbb{Q} = \{a + q \mid q \in \mathbb{Q}\}$. And, given a basis element $(a, b) \in \mathbb{R}$, for the projection $\pi: \mathbb{R} \rightarrow \mathbb{R}/\sim: a \mapsto a + \mathbb{Q}$.

$$\pi^{-1}[\pi[(a, b)]] = \bigcup_{r \in (a, b)} r + \mathbb{Q} = \bigcup_{q \in \mathbb{Q}} (a + q, b + q)$$

is open and since π is quotient map, so $\pi[(a, b)]$ is open, so π is open. Therefore, the topology on $\mathbb{R}/\sim$ is generated by

$$\mathcal{B}_{\sim} = \{\pi[(a, b)] \mid a < b\} = \{(a, b) + \mathbb{Q} \mid a < b\}.$$

But, by the density of rationals, for any $a < b$, $(a, b) + \mathbb{Q} = \mathbb{R}/\mathbb{Q}$. Therefore the topology on $\mathbb{R}/\sim$ is indiscrete.

2025, Fall.

13. Consider a regular plane curve $C: I \rightarrow \mathbb{R}^2$.

a). Give a formula for the normal n to C , and define the signed curvature.

b). Explain the relation between the total angle change and the signed curvature.

c). Assume C is a closed plane curve parametrized by arc-length, and $I = [0, 2\pi]$.

Prove or disprove that: there are $t, t' \in I$ such that $C'(t) = -C'(t')$.

and if $t=0$, there is such a $t' \in I$?

Moreover, show that $\int_0^{2\pi} |K(s)| ds \geq 2\pi$ where $K(s)$ is the signed curvature.

Proof. Denote $C(t) = (x(t), y(t))$.

a). The tangent vector is

$$T(t) = \frac{C'(t)}{|C'(t)|} = \frac{(x'(t), y'(t))}{\sqrt{x'(t)^2 + y'(t)^2}}$$

So, the normal vector is

$$N(t) = \frac{(-y'(t), x'(t))}{\sqrt{x'(t)^2 + y'(t)^2}}$$

And, since the curvature $K(s(t))$ satisfies $T'(t) = K(t) \cdot N(t)$,

$$K(t) = K(t) \langle N(t), N(t) \rangle = \langle T'(t), N(t) \rangle$$

Pebris,

Denote $C = (x, y)$. Then,

$$T = (x', y') / |(x', y')| = \frac{(x', y')}{\sqrt{x'^2 + y'^2}}$$

is the tangent vector, and the normal vector is

$$N = \frac{(-y', x')}{\sqrt{x'^2 + y'^2}}$$

Moreover, the signed curvature K satisfies $K N = T'$,

so $K = K \cdot \langle N, N \rangle = \langle T', N \rangle \cdot |C'(t)|^{-1}$ Meanwhile

$$T' = \frac{(x'', y'')}{\sqrt{x'^2 + y'^2}} + (x', y') \cdot \left(-\frac{1}{2} \cdot \frac{2x'x'' + 2y'y''}{(x'^2 + y'^2)^{\frac{3}{2}}} \right) \text{ so}$$

$$K = \frac{1}{x'^2 + y'^2} \cdot (x'y'' - x''y') \cdot \frac{1}{\sqrt{x'^2 + y'^2}}$$

TSNU. Q. 14.

Prove that $\lim_{b \rightarrow \infty} \int_0^b \frac{\sin x}{x} dx$ exists.

Proof. First, let $a_n = n\pi$ ($n \in \mathbb{N}$). Then, a sequence $\left\{ \int_0^{a_n} \frac{\sin x}{x} dx \right\}$ converges, because:

$\left| \int_{n\pi}^{(n+1)\pi} \frac{\sin x}{x} dx \right|$ is decreasing and converges to zero, and

$\int_{n\pi}^{(n+1)\pi} \frac{\sin x}{x} dx$ is positive if n is even and is negative if n is odd.

That is, $\int_0^{n\pi} \frac{\sin x}{x} dx = \sum_{k=1}^n \int_{a_{k-1}}^{a_k} \frac{\sin x}{x} dx$ is alternating series, so converges.

7226. Prima.

3. Let V be an n -dimensional vector space over $\mathbb{C}$,
Let $T: V \rightarrow V$ be a linear operator such that $T^3 = T$.

27m

a). Find all possible characteristic polynomials and corresponding minimal polynomials for T .

b). Prove $V = \ker T \oplus \operatorname{Im} T$.

Proof.

a). Since $T^3 - T = 0$, the minimal polynomial $m(t)$ for T divides $t^3 - t = 0$.

$t^3 - t = t(t-1)(t+1)$, there are possibilities:

$$m(t) = t \text{ or } t-1 \text{ or } t+1 \text{ or } t(t-1) \text{ or } t(t+1) \text{ or } (t-1)(t+1) \text{ or } t(t-1)(t+1).$$

Since $m(t)$ divides the characteristic polynomial and they have same characteristic values, the possible characteristic polynomial is:

$$t^{r_1} \cdot (t-1)^{r_2} \cdot (t+1)^{r_3} \text{ where } r_1, r_2, r_3 \text{ are non-negative integers and } r_1 + r_2 + r_3 = n.$$

If $r_1 = 0$ and $r_2, r_3 \neq 0$, then the corresponding minimal polynomial is $(t-1)(t+1)$

and other cases are similar.

b). ~~Given $\alpha \in \ker T \cap \operatorname{Im} T$, $T\alpha = 0$ and $\alpha = T\beta$ for some $\beta \in V$.~~

Given $\alpha \in V$, $\alpha = T^2\alpha + \alpha - T^2\alpha$ and since $T(\alpha - T^2\alpha) = (T - T^3)\alpha = 0\alpha = 0$,
so $\alpha - T^2\alpha \in \ker T$ and trivially $T^2\alpha \in \operatorname{Im} T$. Moreover if $T\alpha = 0$ and $\alpha \in \operatorname{Im} T$,
Then $\alpha = T\beta = T^3\beta$ for some $\beta \in V$ and $T\alpha = T^2\beta = 0$ so $T^3\beta = TT^2\beta = \alpha = 0$,
so $V = \ker T \oplus \operatorname{Im} T$.

Modified.

Let T be a linear operator on the fin. dim space V over $\mathbb{C}$ satisfying $T^4 = T^2$.

a). Prove that $V = \ker T^2 \oplus \operatorname{Im} T^2$.

b). Describe $T|_{\operatorname{Im} T^2}$.

c). Prove that T is diagonalizable $\Leftrightarrow \ker T = \ker T^2$.

observe: $T^4 - T^2 = 0$, so the minimal polynomial divides $t^2(t-1)(t+1)$.

a). Given $\alpha \in V$, $\alpha = T^2\alpha + \alpha - T^2\alpha$ and $T^2\alpha \in \operatorname{Im} T^2$ clearly, and $T^2(I - T^2)\alpha = (T^2 - T^4)\alpha = 0 \cdot \alpha = 0$, so $\alpha - T^2\alpha$, which proves $\alpha - T^2\alpha \in \ker T^2$, so $V = \ker T^2 + \operatorname{Im} T^2$. Moreover, given $\alpha \in \ker T^2 \cap \operatorname{Im} T^2$, $T^2\alpha = 0$ and $\alpha = T^2\beta$ for some $\beta \in V$, so

$$\alpha = T^2\beta = T^4\beta = T^2T^2\beta = T^2\alpha = 0$$

so $V = \ker T^2 \oplus \operatorname{Im} T^2$.

b). Given $\alpha \in \operatorname{Im} T^2$, $\alpha = T^2\beta$ for some $\beta \in V$. so, $T\alpha = T^3\beta$.

Multiplying T , then $T^2\alpha = T^4\beta = T^2\beta$, so $T^2(\alpha - \beta) = 0$.

Therefore, $\alpha - \beta \in \ker T^2$, but since $\alpha \in \operatorname{Im} T^2$ and $\operatorname{Im} T^2 \cap \ker T^2 = 0$,

$\beta = \alpha + \gamma$, where $\gamma \in \ker T^2$. Now, $T\alpha = T^3\beta = T^3\alpha + T^3\gamma = T^3\alpha$,

so, $T|_{\operatorname{Im} T^2} = T^3|_{\operatorname{Im} T^2}$, moreover, since $T^2 = T^4$, $T^2|_{\operatorname{Im} T^2} = I$.

c). Since T is diagonalizable $\Leftrightarrow$ minimal polynomial of T splits and separable.

That is, the minimal polynomial is $t^a(t-1)^b(t+1)^c$, $a, b, c \in \{0, 1\}$.

2026 Spring

5. Let $f(x) = x^4 + 2x + 3 \in \mathbb{F}_7[x]$ and $K = \mathbb{F}_7[x]/(f)$.

Let $\mathcal{Q}: K \rightarrow K: a \mapsto a^7$.

a). Prove K is a field and $\mathcal{Q}$ is Automorphism.

b). Let $G = \langle \mathcal{Q} \rangle \leq \text{Aut}(K)$. Determine the number of $a \in K$ such that $|\text{Orb}(a)| = 2$.

Proof

a). Since $f(0)=3, f(1)=6, f(2)=2, f(3)=6, f(4)=1, f(5)=1, f(6)=2$, so $f(x)$ has no linear factor. So, if $f(x)$ is reducible,

$$f(x) = (x^2 + ax + b)(x^2 + cx + d) \text{ for some } a, b, c, d \in \mathbb{F}_7.$$

Then,
$$\begin{cases} a+c=0 & \text{If } a=0 \text{ or } c=0, \text{ then } a=c=0, \text{ so } 0=2. \\ ac+b+d=0 & \text{If } b=0 \text{ or } d=0, \text{ then } 0=3, \\ ad+bc=2 & \text{Therefore, } a, b, c, d \text{ are non-zero.} \\ bd=3 & \text{Reduce the system:} \end{cases}$$

$$\Rightarrow \begin{cases} b+d=a^2 \\ a(d-b)=2 \\ bd=3 \end{cases}$$

If $b=d$, then $b^2=3$, but $\sqrt{3} \notin \mathbb{F}_7$, so $b \neq d$.

And, $3 = 1 \cdot 3$ or $2 \cdot 5$ or $4 \cdot 6$, so

$a^2 = 4$ or 0 or 3 . Since $\sqrt{3} \notin \mathbb{F}_7$ and $a \neq 0$, $a^2 = 4$, so $a = 2$ or 5 ,

with $\{b, d\} = \{1, 3\}$. But, $2 \cdot (1-3) = 3 \neq 2$,
 $2 \cdot (3-1) = 4 \neq 2$,
 $5 \cdot (1-3) = 4 \neq 2$,
 $5 \cdot (3-1) = 3 \neq 2$

So contradiction. Therefore, f must be an irreducible.

Since $\mathbb{F}_7$ is a field, $\mathbb{F}_7[x]$ is p.z.d., so (f) is maximal, so $\mathbb{F}_7[x]/(f)$ is a field. And, since the characteristic of K is 7, for any $a, b \in K$,

$$(a+b)^7 = \sum_{k=0}^7 \binom{7}{k} a^{7-k} b^k = a^7 + b^7$$

and $(ab)^7 = a^7 b^7$, so $\mathcal{Q}$ is Homomorphism. Moreover,

$\mathcal{Q}(a) = 0 \Leftrightarrow a^7 = 0 \Leftrightarrow a = 0$, so $\ker \mathcal{Q}$ is trivial, so $\mathcal{Q}$ is 1-1.

Now, since $[K: \mathbb{F}_7] = \deg f = 4$, so $\#K = 7^4$, finite, so $\mathcal{Q}$ is bijective.

b). Since every finite field is perfect, so K is Galois over $\mathbb{F}_7$,
so $|\text{Aut}(K)| = \deg f = 4$,

Consider an Action

$$G \times K \rightarrow K : (\mathbb{Q}^n, \alpha) \mapsto \mathbb{Q}^n(\alpha).$$

So $|\text{Orb}(\alpha)| = 2 \iff \mathbb{Q}^2(\alpha) = \mathbb{Q}$ and $\alpha \neq 0 \iff \alpha^{4q} = \alpha$ and $\alpha \neq 0$
So, the fixed field by $\langle \mathbb{Q}^2 \rangle$, $\langle \mathbb{Q}^2 \rangle \setminus \{0\}$ is we desired.

Revised in the next

Revised.

Prove that $f(x) = x^4 + 2x + 3$ is irreducible over $\mathbb{F}_7$.

And describe the Galois group of $f(x)$.

1). $f(x)$ has no linear factor.

$f(0) = 3 \neq 0$. And, since $\mathbb{F}_7^\times$ is a group of order 6 under the multiplication. Therefore, for any $\alpha \in \mathbb{F}_7^\times$, $\alpha^6 = (\alpha^3)^2 = 1$ implies $\alpha^3 = \pm 1$.

If $\alpha^3 = 1$, then $\alpha^4 = \alpha$, so $f(\alpha) = \alpha^4 + 2\alpha + 3 = 3(\alpha + 1) = 0$ implies $\alpha = -1$ but $(-1)^3 = -1 = 6 \neq 1$, so $\alpha^3 \neq 1$.

If $\alpha^3 = -1$, then $\alpha^4 = -\alpha$, so $f(\alpha) = \alpha + 3 = 0$ implies $\alpha = 4$, but $4^3 = 64 = 1 \neq -1$, so $\alpha^3 \neq -1$. Now, $f(x)$ has no root in $\mathbb{F}_7^\times$.

2). $f(x)$ has no factor of degree 2.

Suppose that $f(x) = (x^2 + ax + b)(x^2 + cx + d)$. Then, we have a system:

$$\begin{cases} a+c=0 \\ ac+b+d=0 \\ ad+bc=2 \\ bd=3 \end{cases} \Rightarrow \begin{cases} b+d=a^2 \\ a(d-b)=2 \\ bd=3 \end{cases} \Rightarrow \begin{cases} (d-b)^2 = (b+d)^2 - 4bd = a^4 - 12 = a^4 + 2 \\ a^2(d-b)^2 = 4 \end{cases}$$

$\alpha \neq 0$, so $\alpha^6 = 1$

$\Rightarrow \alpha^6 + 2\alpha^2 = 1 + 2\alpha^2 = 4 \Rightarrow \alpha^2 = 5$. But, $\{x^2 \mid x \in \mathbb{F}_7\} = \{0, 1, 4, 2\}$, so $\sqrt{\alpha} \notin \mathbb{F}_7$.

Now, f is irreducible $\Rightarrow \mathbb{F}_7[x]/(f)$ is a field.

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2. Let $A \in \text{Mat}_4(\mathbb{R})$ be a symmetric with eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \lambda_4$.

Let
$$P = \frac{1}{2} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

a) Prove that P is an orthogonal projection.
 b) prove $\text{tr}(AP) \leq \lambda_1 + \lambda_2$
 c) $\max_{\substack{x, y \in \mathbb{R}^4 \\ \|x\| = \|y\| = 1 \\ \langle x, y \rangle = 0}} x^T A x + y^T A y = \lambda_1 + \lambda_2$.

a). By just calculation, $P^2 = P = P^T$, and in $\mathbb{R}^4$, $P^T = P^*$, so P is an orthogonal projection, $\text{rank } P = 2$, because the row vectors space has dimension 2.

b). Since P is orthogonal projection, $\mathbb{R}^4 = \text{Im } P \oplus \text{Ker } P$, with $(\text{Ker } P)^\perp = \text{Im } P$, and $\text{rank } P = \text{nullity } P = 2$. Since A is symmetric, there are four eigenvectors $\{\alpha_1, \dots, \alpha_4\}$ corresponding to $A\alpha_i = \lambda_i \alpha_i$ ($i=1, \dots, 4$) and $\{\alpha_1, \dots, \alpha_4\}$ is orthonormal.

And, let $\{v_1, v_2\}$ be a basis for $\text{Im } P$ and $\{v_3, v_4\}$ be a basis for $\text{Ker } P$, satisfying $\{v_1, \dots, v_4\}$ is orthonormal.

Since the trace is invariant under the similar, for $B = (v_1 \dots v_4)$,

$$\begin{aligned} \text{tr}(AP) &= \text{tr}(B^T A P B) = v_1^T A P v_1 + \dots + v_4^T A P v_4 = v_1^T A v_1 + v_2^T A v_2 \\ &= \langle A v_1, v_1 \rangle + \langle A v_2, v_2 \rangle \end{aligned}$$

Now, since $v_1, v_2 \in \langle \alpha_1, \dots, \alpha_4 \rangle$ and $\|v_1\| = \|v_2\| = 1$ and $\langle v_1, v_2 \rangle = 0$, $v_1 = a_1 \alpha_1 + \dots + a_4 \alpha_4$ and $v_2 = b_1 \alpha_1 + \dots + b_4 \alpha_4$ such that $a_1^2 + \dots + a_4^2 = b_1^2 + \dots + b_4^2 = 1$ and $a_1 b_1 + \dots + a_4 b_4 = 0$. Now,

$$\begin{aligned} \langle A v_1, v_1 \rangle + \langle A v_2, v_2 \rangle &= a_1^2 \lambda_1 + \dots + a_4^2 \lambda_4 + b_1^2 \lambda_1 + \dots + b_4^2 \lambda_4 \\ &= \lambda_1 (a_1^2 + b_1^2) + \dots + \lambda_4 (a_4^2 + b_4^2) \end{aligned}$$

But, for each $1 \leq i \leq 4$, $a_i^2 + b_i^2 \leq 1$ and $a_1^2 + \dots + a_4^2 + b_1^2 + \dots + b_4^2 = 2$, therefore $\leq \lambda_1 \cdot 1 + \lambda_2 \cdot 1 = \lambda_1 + \lambda_2$. It proves b)

c) Given $x, y \in \mathbb{R}^4$ such that $\langle x, y \rangle = 0$, with $\|x\| = \|y\| = 1$, $x^T A x + y^T A y = \langle A x, x \rangle + \langle A y, y \rangle \leq \lambda_1 + \lambda_2$, by the argument in b). And, take $x = \alpha_1$ and $y = \alpha_2$, we obtain c).

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4. Let G be a group and $H_1, H_2 \leq G$ be subgroups of finite index.

a). Prove that the map

$$T: G \rightarrow S_{G/H_1 \times G/H_2} : x \mapsto T_x \text{ where } T_x(aH_1, bH_2) = (xaH_1, xbH_2)$$

is Homomorphism and $\text{Ker } T \subseteq H_1 \cap H_2$.

b). Prove $H_1 \cap H_2$ has finite index.

Proof

$$\begin{aligned} \text{a). Given } g_1, g_2 \in G, T(g_1 g_2)(aH_1, bH_2) &= (g_1 g_2 aH_1, g_1 g_2 bH_2) \\ &= T(g_1) \circ T(g_2)(aH_1, bH_2) \end{aligned}$$

for each $a, b \in G$, so it preserves the operation.

$$\text{And } g \in \text{Ker } T \Leftrightarrow T_g = \text{id} \Leftrightarrow \text{for all } a, b \in G, T_g(aH_1, bH_2) = (gaH_1, gbH_2) = (aH_1, bH_2)$$

$$\begin{aligned} &\Leftrightarrow \text{for all } a, b \in G, a^{-1}ga \in H_1 \text{ and } b^{-1}gb \in H_2 \\ &\Rightarrow g \in H_1 \cap H_2 \text{ (take } a=b=1_G) \end{aligned}$$

b). By the first isomorphism theorem,

$$G/\text{Ker } T \cong \text{Im } T \leq S_{G/H_1 \times G/H_2}$$

and since H_1, H_2 have finite index, so the Codomain of T is finite set.

$$\text{Now, } |G : H_1 \cap H_2| \leq |G : \text{Ker } T| \leq \# S_{G/H_1 \times G/H_2}$$

because $\text{Ker } T \subseteq H_1 \cap H_2$, so proved.

Note: If $A \leq B \leq G$ and $|G:A|$ is finite, then

$$G/A \rightarrow G/B : gA \mapsto gB$$

is well-defined and onto, because: Since $A \leq B$,

$$g_1 A = g_2 A \Leftrightarrow g_1 g_2^{-1} \in A \leq B \Rightarrow g_1 g_2^{-1} \in B \Leftrightarrow g_1 B = g_2 B$$

and surjectiveness is clear.

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8. For $f \in C([0,1])$, define its extension

$$\tilde{f}(x) := \begin{cases} f(0) & \text{if } x < 0 \\ f(x) & \text{if } 0 \leq x \leq 1 \\ f(1) & \text{if } x > 1 \end{cases}$$

For each $n \geq 1$, define

$$p_n(x) := \frac{n}{2} \cdot \chi_{[-\frac{1}{n}, \frac{1}{n}]}(x)$$

Define $T_n f(x) := \int_{\mathbb{R}} p_n(x-y) \tilde{f}(y) dy$, $x \in \mathbb{R}$.

a). Show that $T_n f \rightarrow f$ on $[0,1]$.

b). Determine $T_n f \rightarrow f$ uniformly on $[0,1]$ or not.

Proof. Given $x \in \mathbb{R}$ and $n \in \mathbb{N}$, $x-y \in [-\frac{1}{n}, \frac{1}{n}] \Leftrightarrow -\frac{1}{n} + x \leq y \leq \frac{1}{n} + x$. Therefore,

$$T_n f(x) = \int_{-\frac{1}{n}+x}^{\frac{1}{n}+x} \frac{n}{2} \cdot \tilde{f}(y) dy.$$

Since $\tilde{f}$ is continuous on $[-1,2]$, so it is continuously uniformly.

Given $\varepsilon > 0$, take $\delta > 0$ such that $\forall x, t \in [0,1]$, $|x-t| < \delta \Rightarrow |f(x) - f(t)| < \varepsilon$.

Now, given $x \in [0,1]$, and $n > \frac{1}{\delta}$,

$$\begin{aligned} |T_n f(x) - f(x)| &= \left| \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} \frac{n}{2} \tilde{f}(y) dy - \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} \frac{n}{2} f(x) dy \right| \\ &= \left| \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} \frac{n}{2} [\tilde{f}(y) - f(x)] dy \right| \\ &\leq \frac{n}{2} \cdot \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} |\tilde{f}(y) - f(x)| dy \\ &< \frac{n}{2} \cdot \frac{2}{n} \cdot \varepsilon = \varepsilon \end{aligned}$$

Therefore, $\sup_{x \in [0,1]} |T_n f - f| \rightarrow 0$ as $n \rightarrow \infty$, so converges uniformly.



